ALGONQUIN COLLEGE

CST8507: NATURAL LANGUAGE PROCESSING

WEEK # 10
BERT &
INTRODUCTION TO
QUESTION/ANSWERING

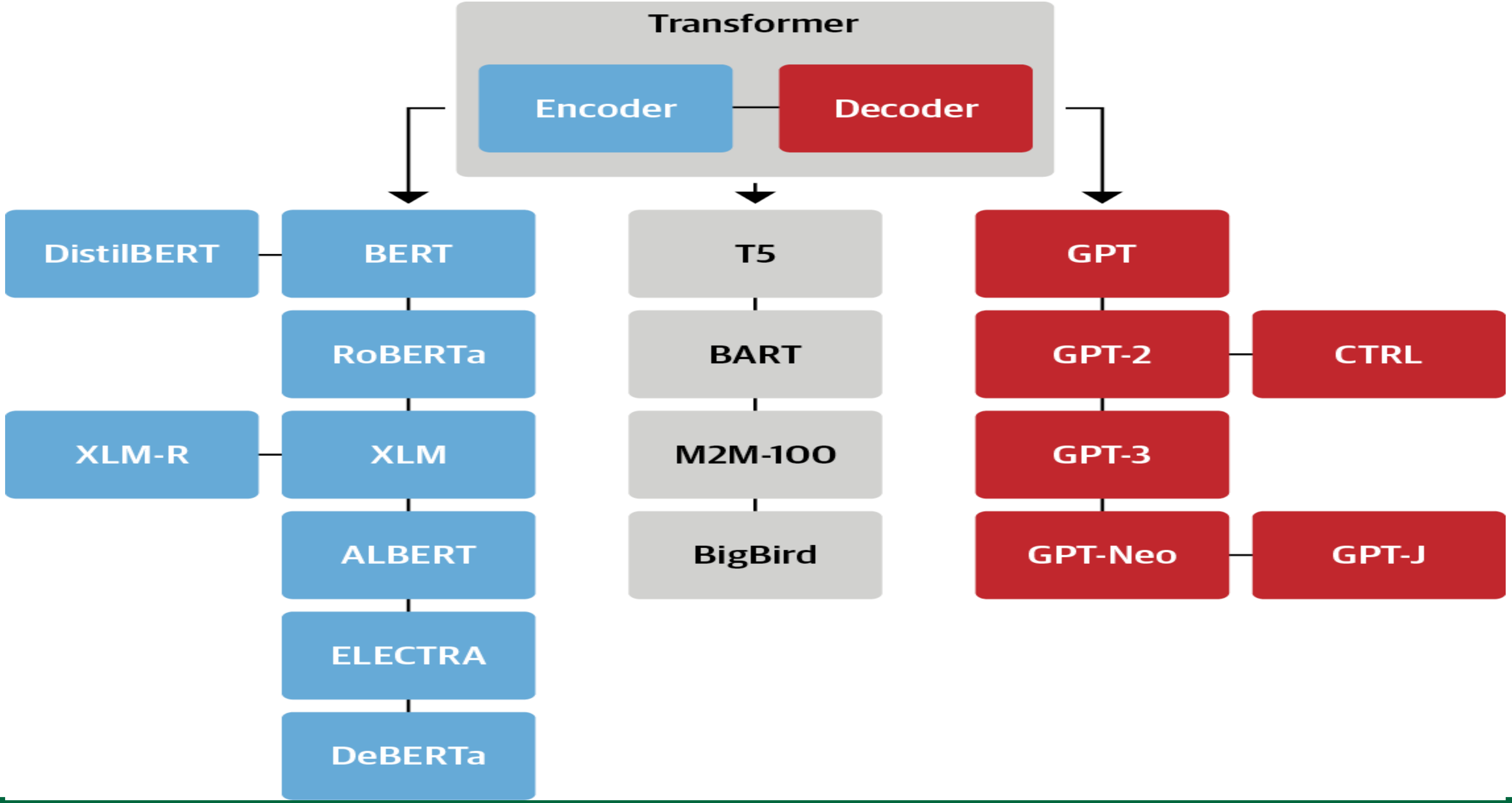
DEVELOPED BY

HALA OWN, PH.D.

Lesson Agenda

- ❑ BERT Architecture
    - BERT VARIANTS
  - BERT FOR TEXT CLASSIFICATION
- ❑ What is Question Answering?
- ❑ Extractive Question Answering (Reading Comprehension)
- ❑ Open Domain Question Answering
- ❑ Closed Domain Question Answering
- ❑ Generative Question Answering

The Transformer Tree of Life



Transformer TimeLine

"BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding" Jacob Devlin et al.

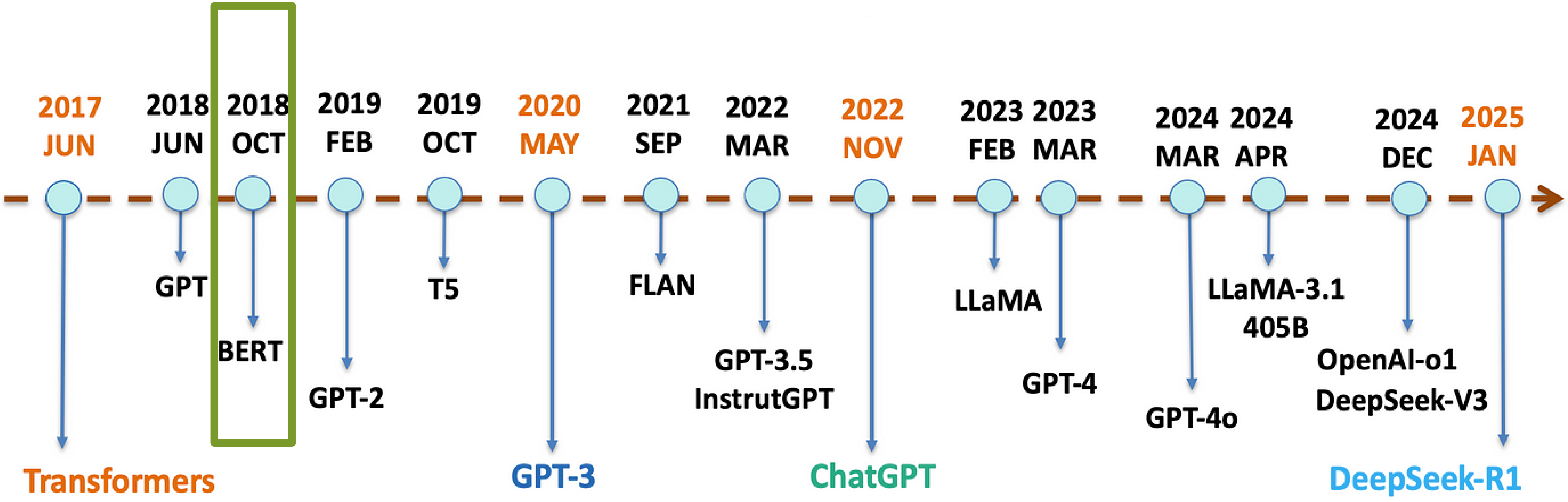
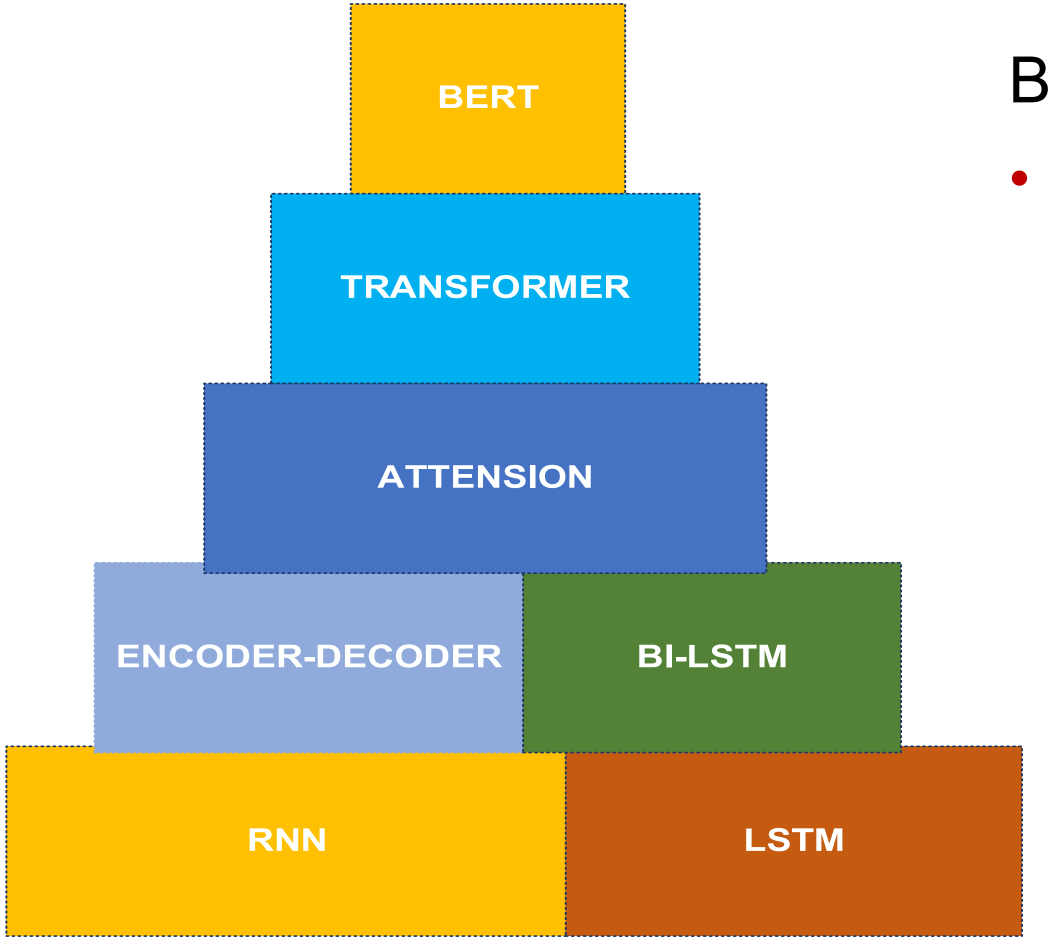


Image source:https://medium.com/@lmpo/a-brief-history-of-lmms-from-transformers-2017-to-deepseek-r1-2025-dae75dd3f59a

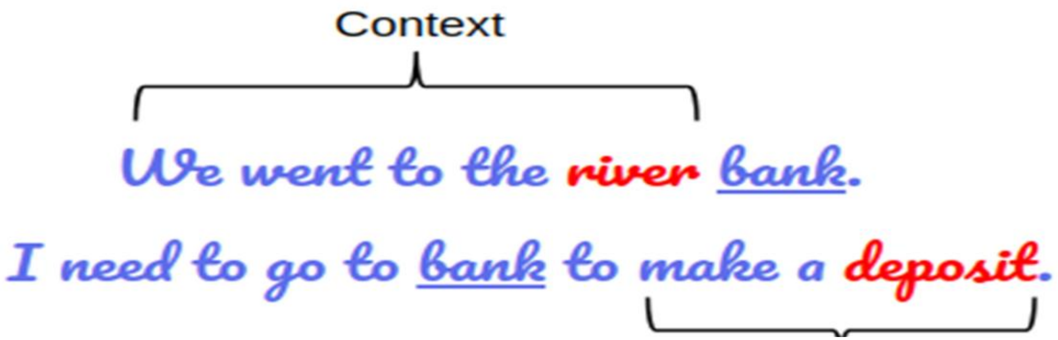
Bidirectional Encoder Representations from Transformers(BERT)

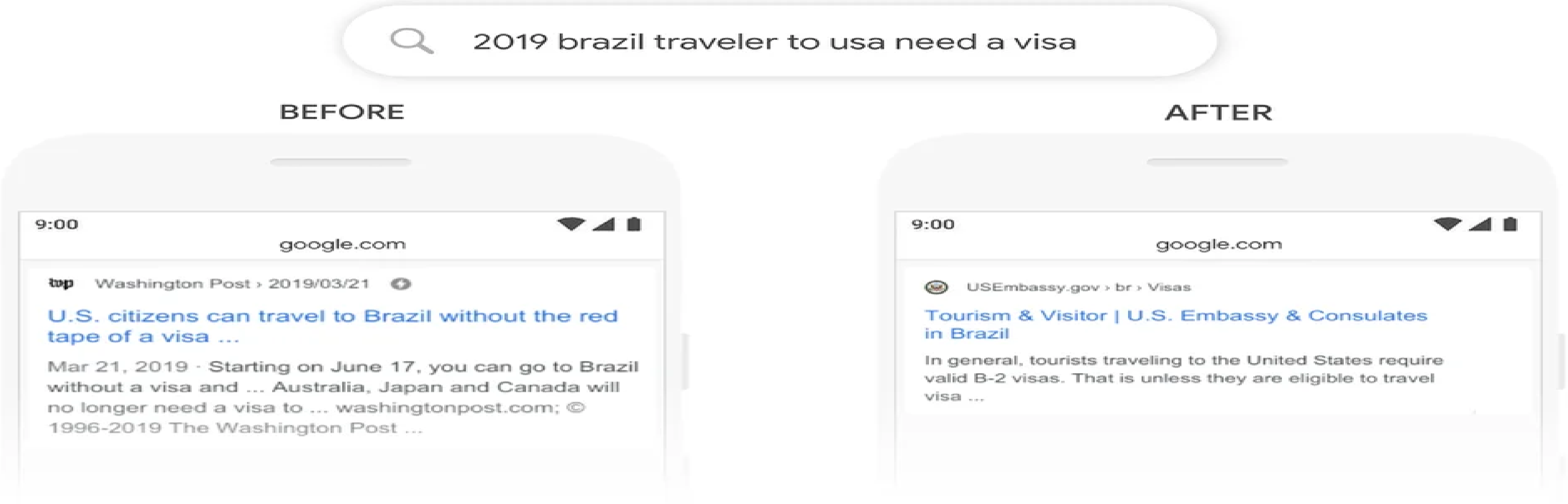


BERT was trained on

- **Data**: English Wikipedia, around 2.5 billion words, Book Corpus, which are 11,000 books around 800 million words.

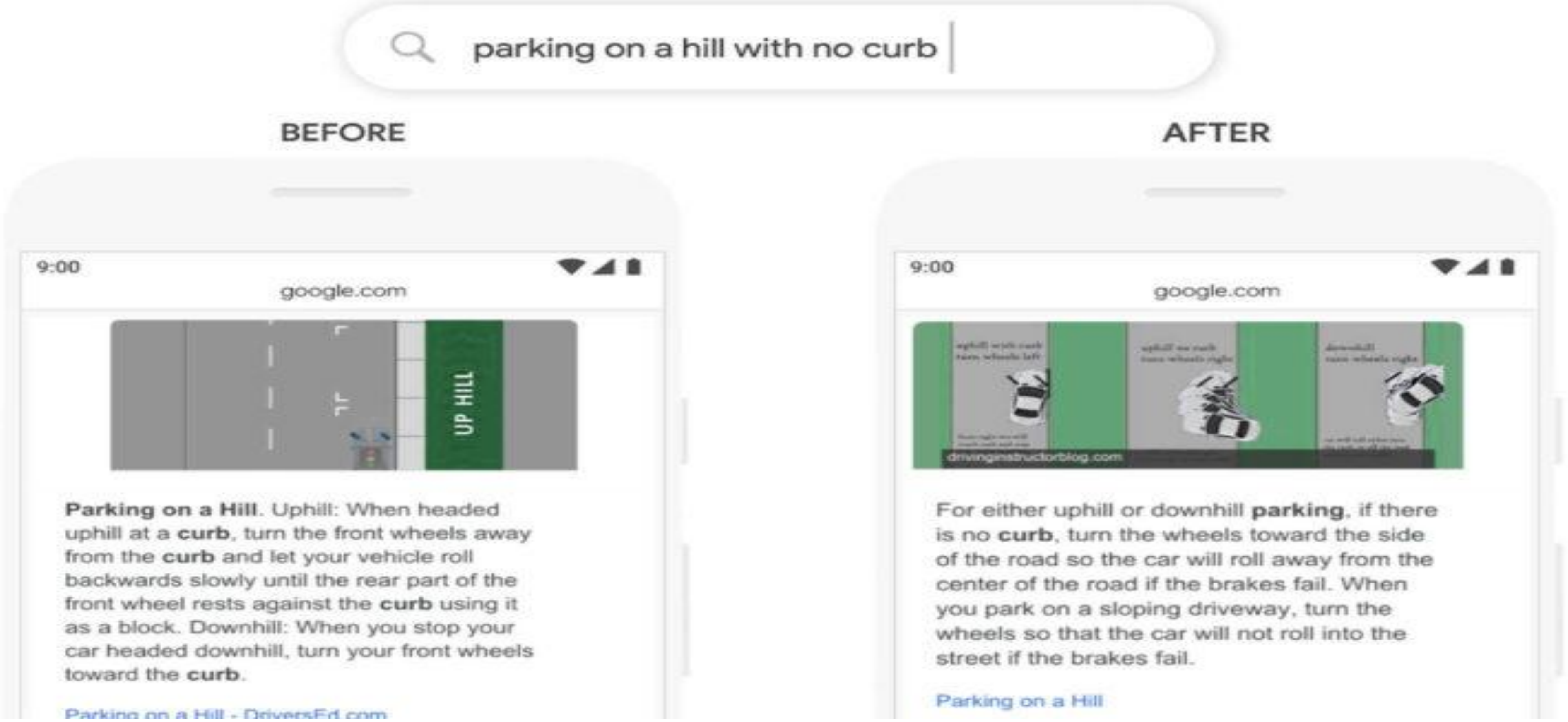
BERT: Google Search





http://jalammar.github.io/a-visual-guide-to-using-bert-for-the-first-time/

BERT: Google Search…



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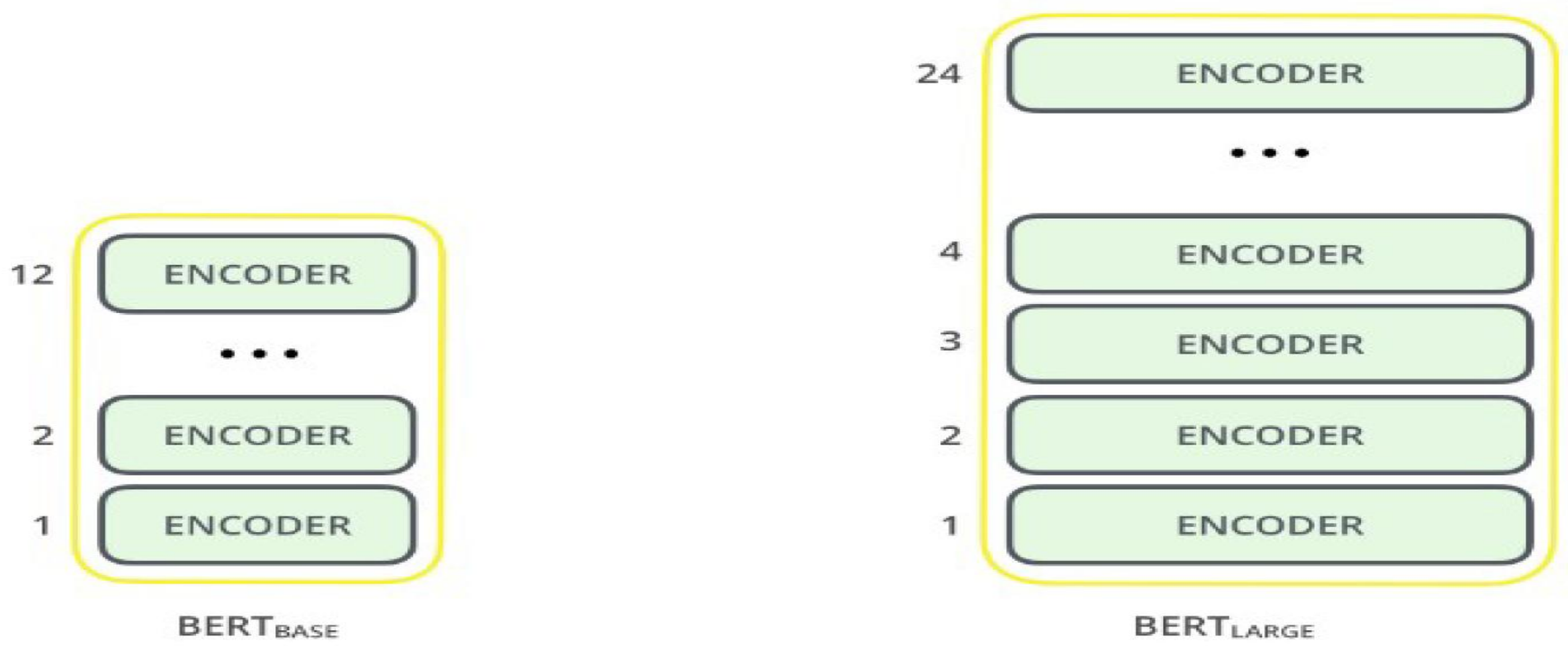
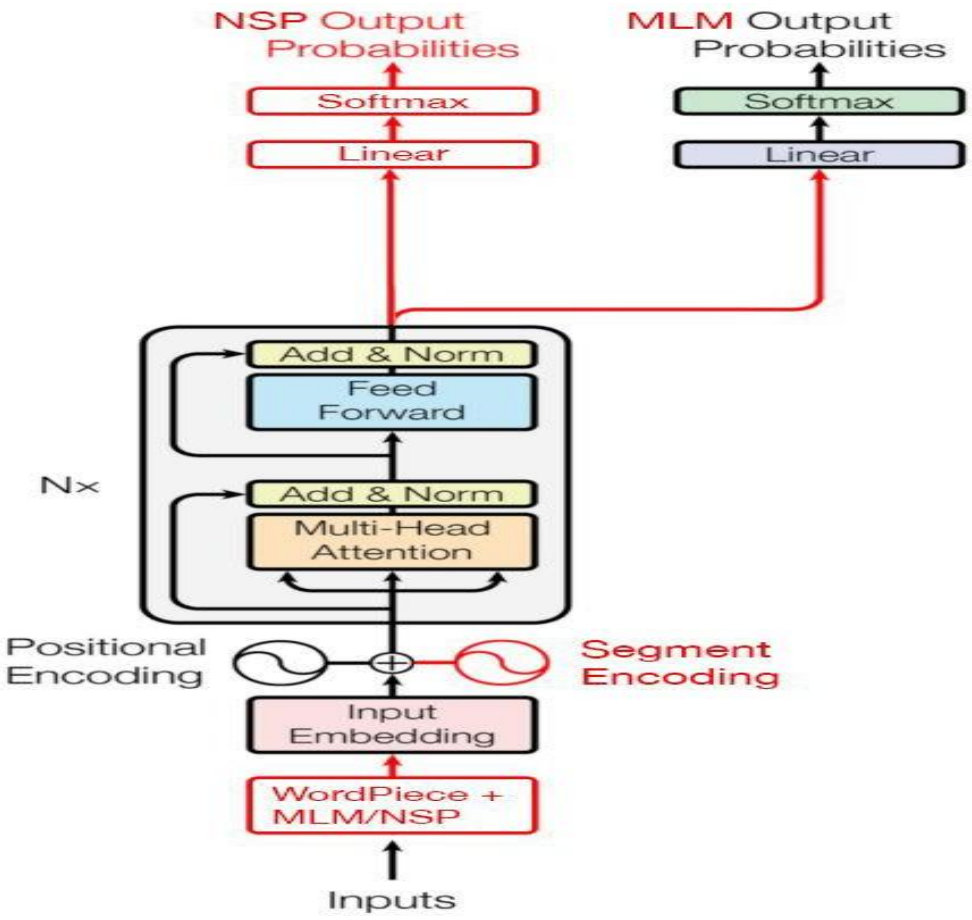
BERT Architecture



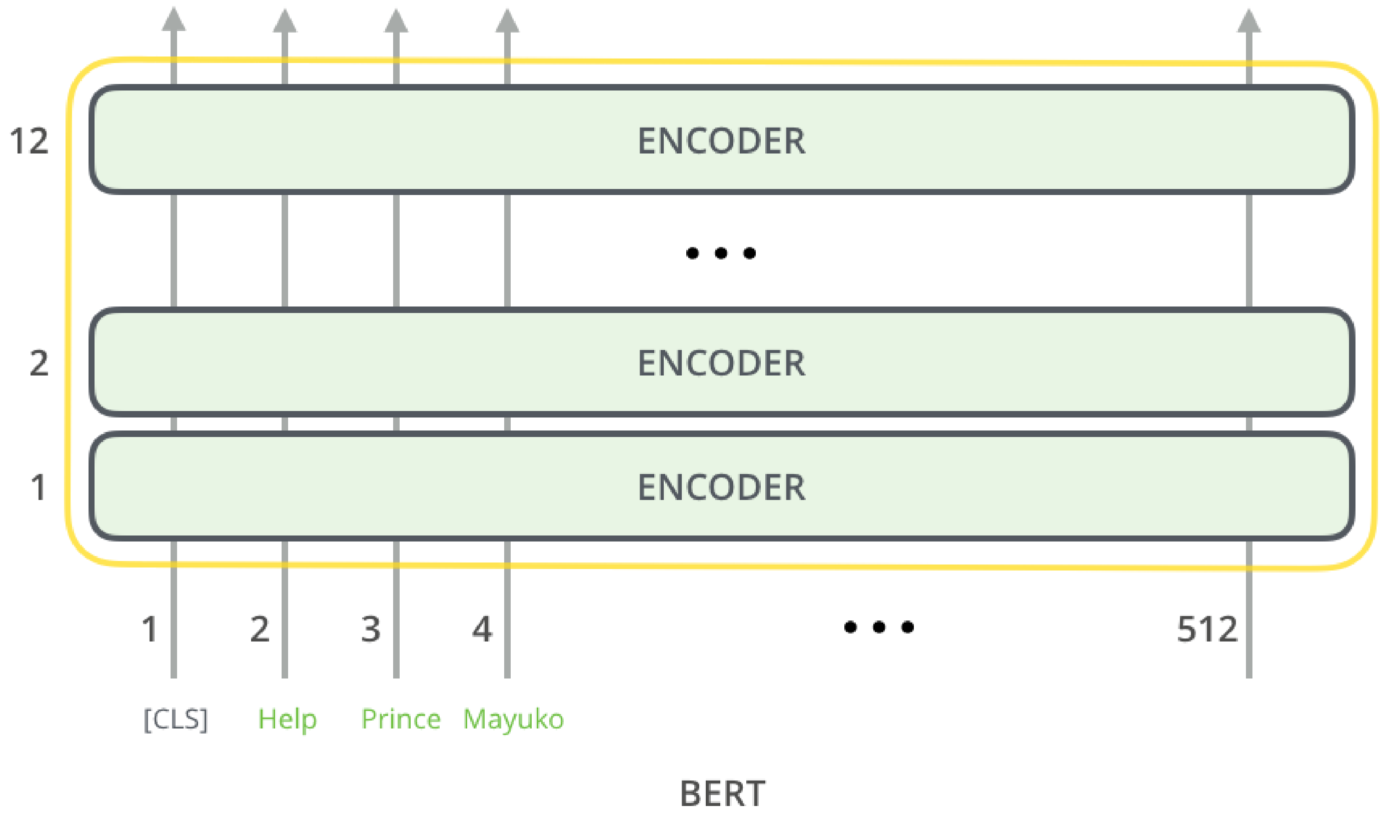
BERT-base: **12** layers, **768** hidden size, **12** attention heads, **110M** parameters

BERT-large: **24** layers, **1024** hidden size, **16** attention heads, **340M** parameters

BERT Architecture …



Model Input



12
ENCODER
• • •
2
ENCODER
1
ENCODER
1
2
3
4
• • •
512
[CLS]
Help
Prince
Mayuko
BERT

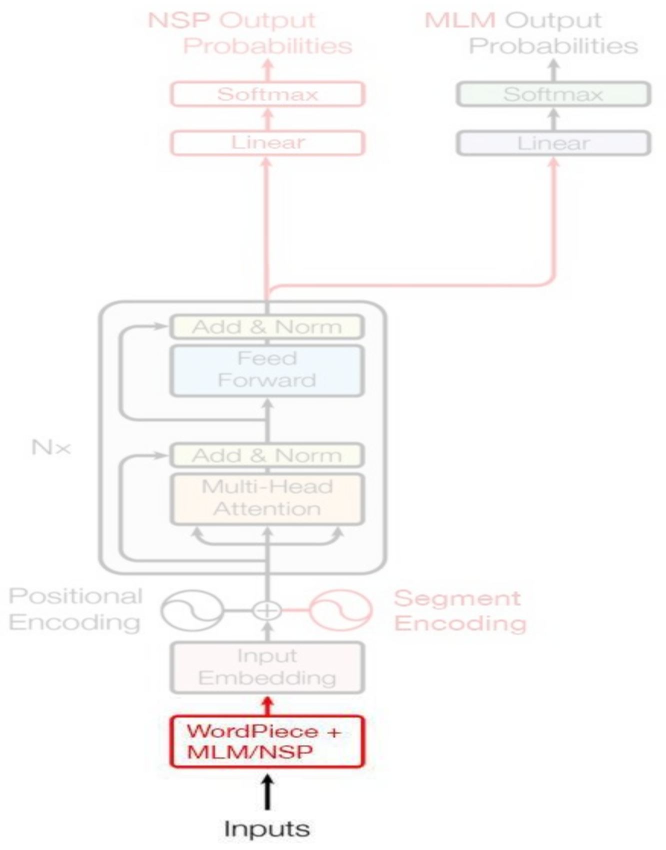
Input processing

WordPiece algorithm

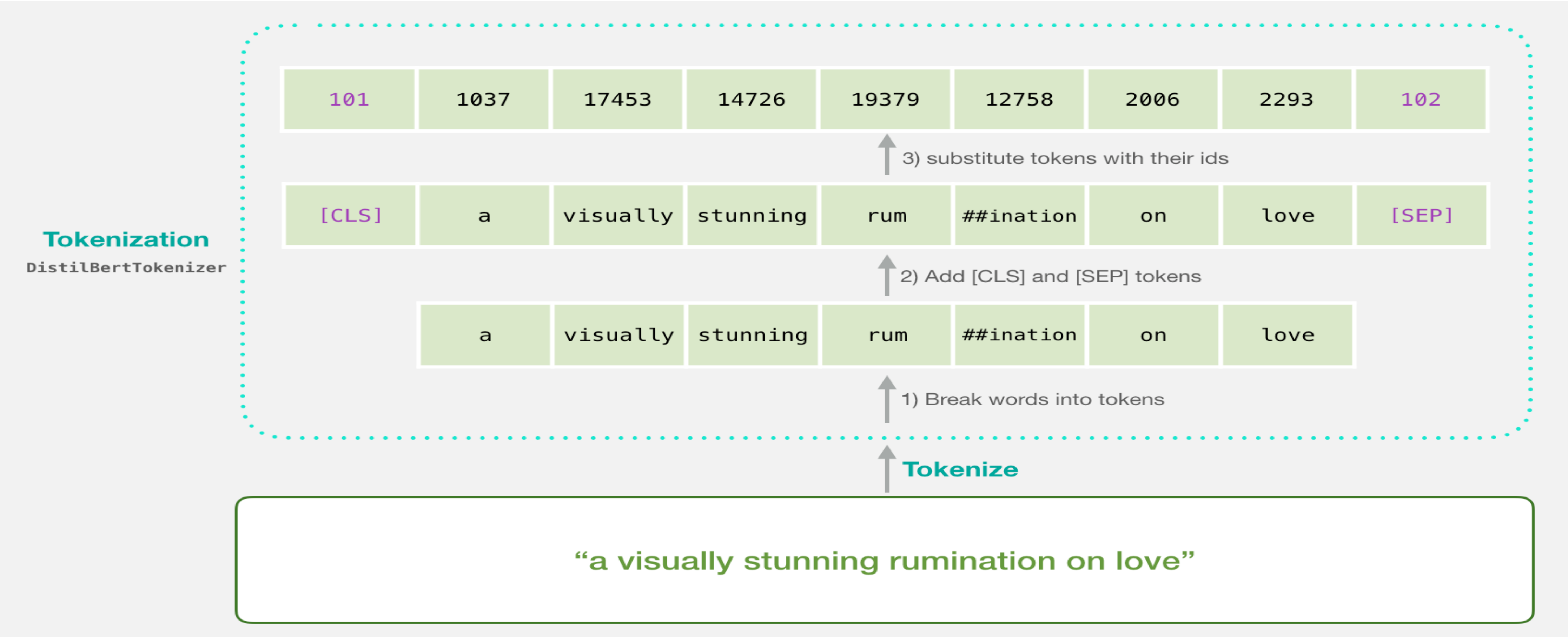
- Tokenizer trained on a training set beforehand
- Vocabulary size: ~30,500

Addind new Tokens

- Add [CLS] token at the beginning of the input
- Separate consecutive segments with the [SEP] token and put another one at the end
- Use [MASK] to mask inputs



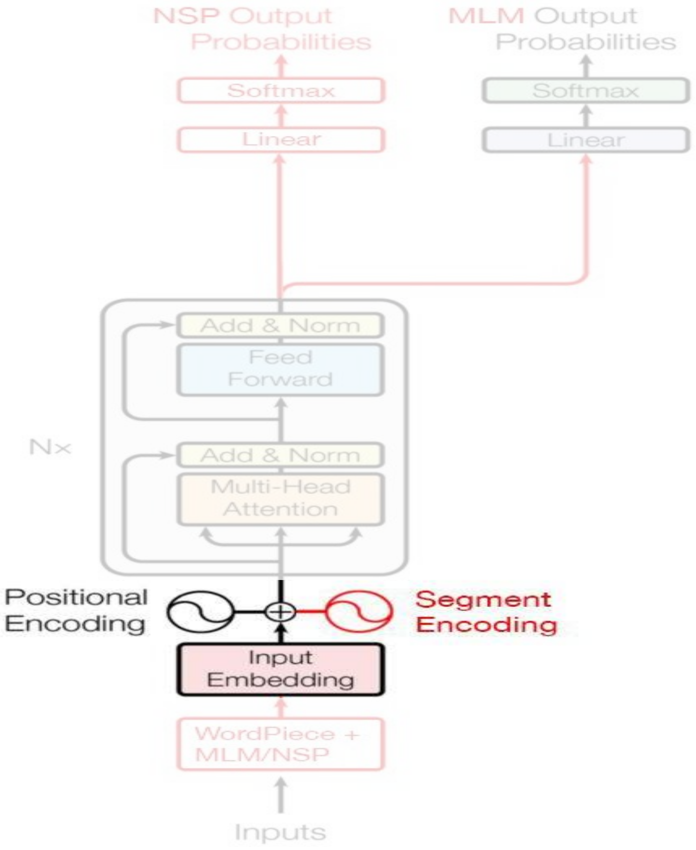
Bert: Tokenization



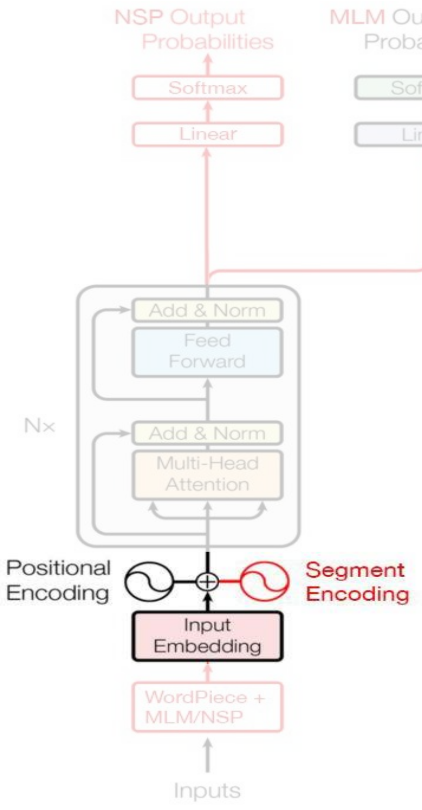
http://jalammar.github.io/a-visual-guide-to-using-bert-for-the-first-time/

Token embedding

[PAD] → 0 [UNK] → 100 [CLS] → 101 [SEP] → 102 [MASK] → 103



BERT: Input Embedding



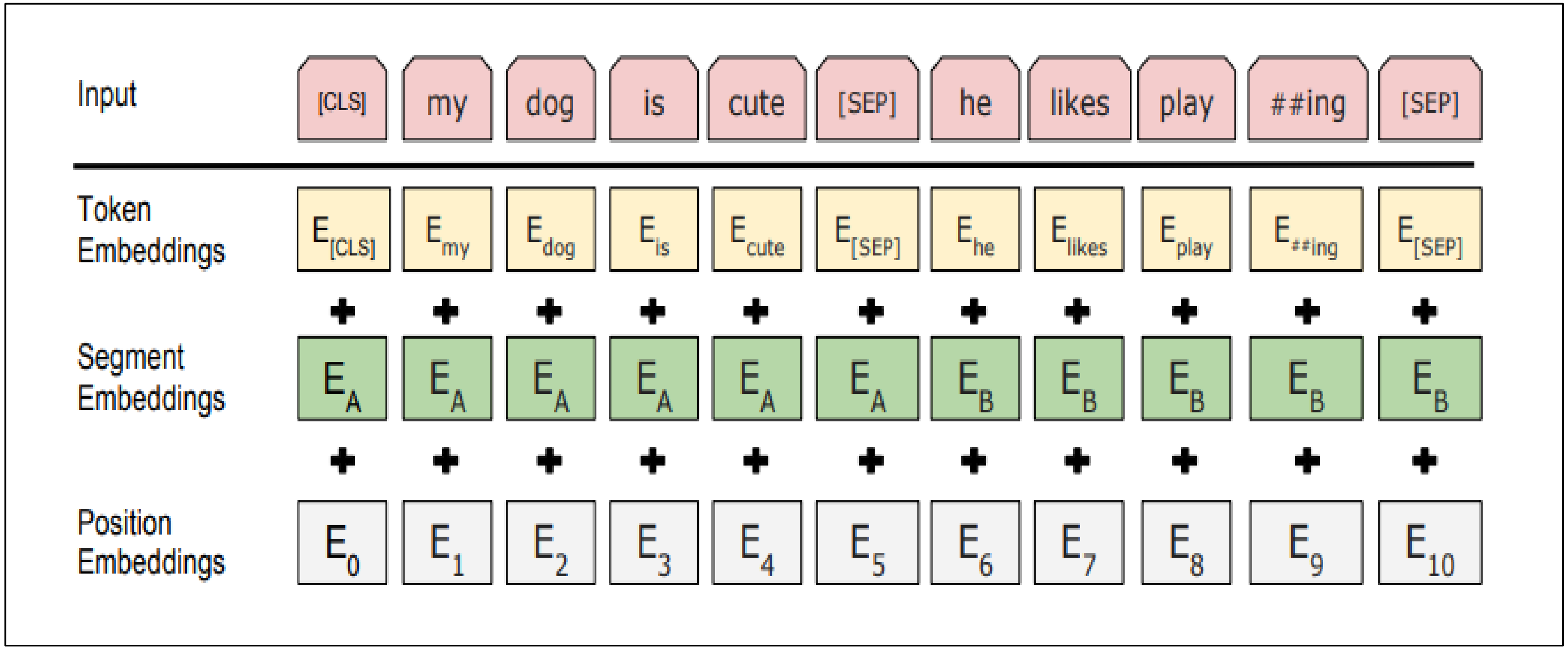
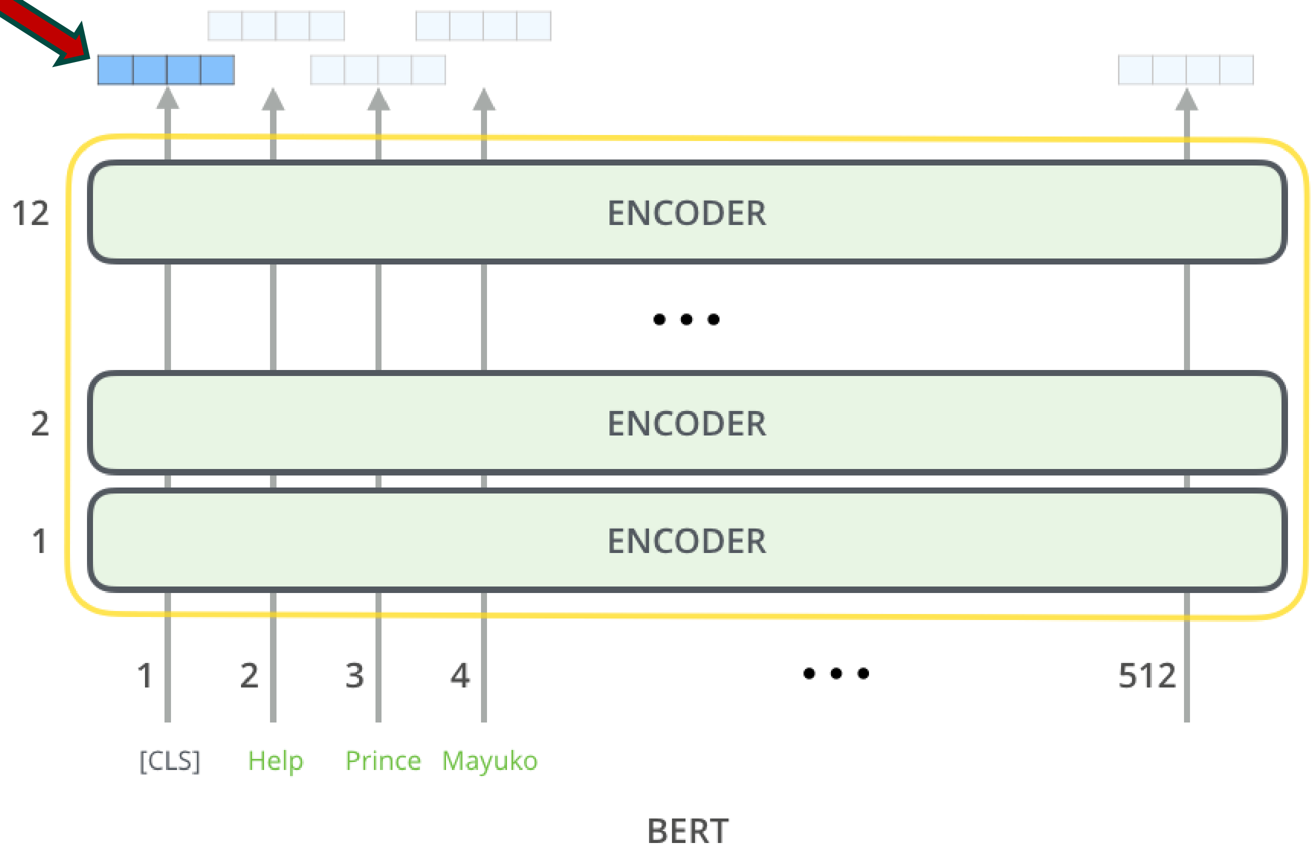


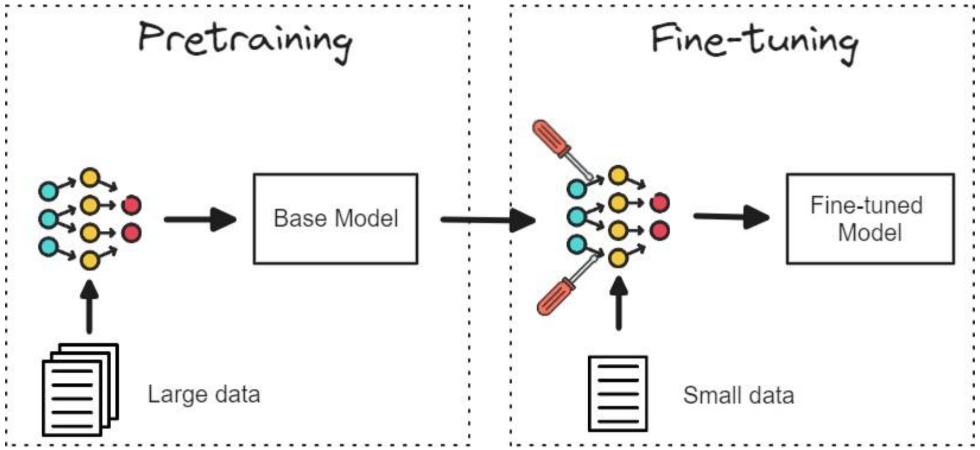
Image adapted from Jacob Devlin, Stanford CS224N)

Model Outputs

[CLS] token after processing Used as **representation for the entire input sentence**



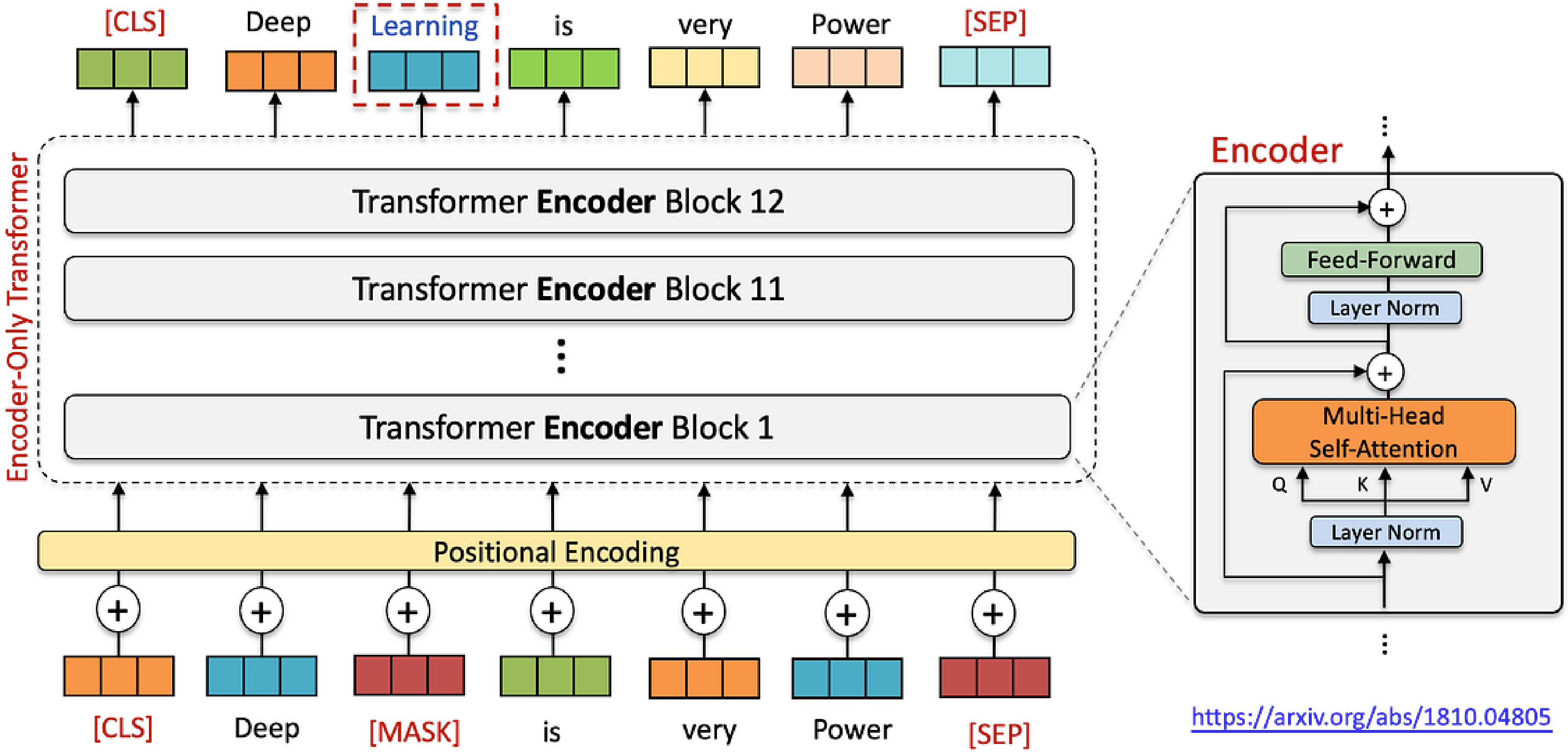
How was Bert Trained



- It happens in two main phases: **pre-training** and **fine-tuning**.
- In **pre-training** phase, BERT learns from a massive amount of text data **without any specific task** in mind.
- In **fine-tuning** phase, we take the pre-trained BERT model and add a small layer on top tailored to a specific task:
    - Masked Language Modelling(MLM)
    - Next Sentence Prediction (NSP)

How was Bert Trained: Masked LM(MLM)

BERT



https://arxiv.org/abs/1810.04805

https://medium.com/@lmpo/bert-unleashed-the-model-that-redefined-language-understanding-afecf5545295

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Masked LM(MLM)

15% of the words to predict.

- 80% of the time, replace with [MASK]

  `went to the store → went to the [MASK]`

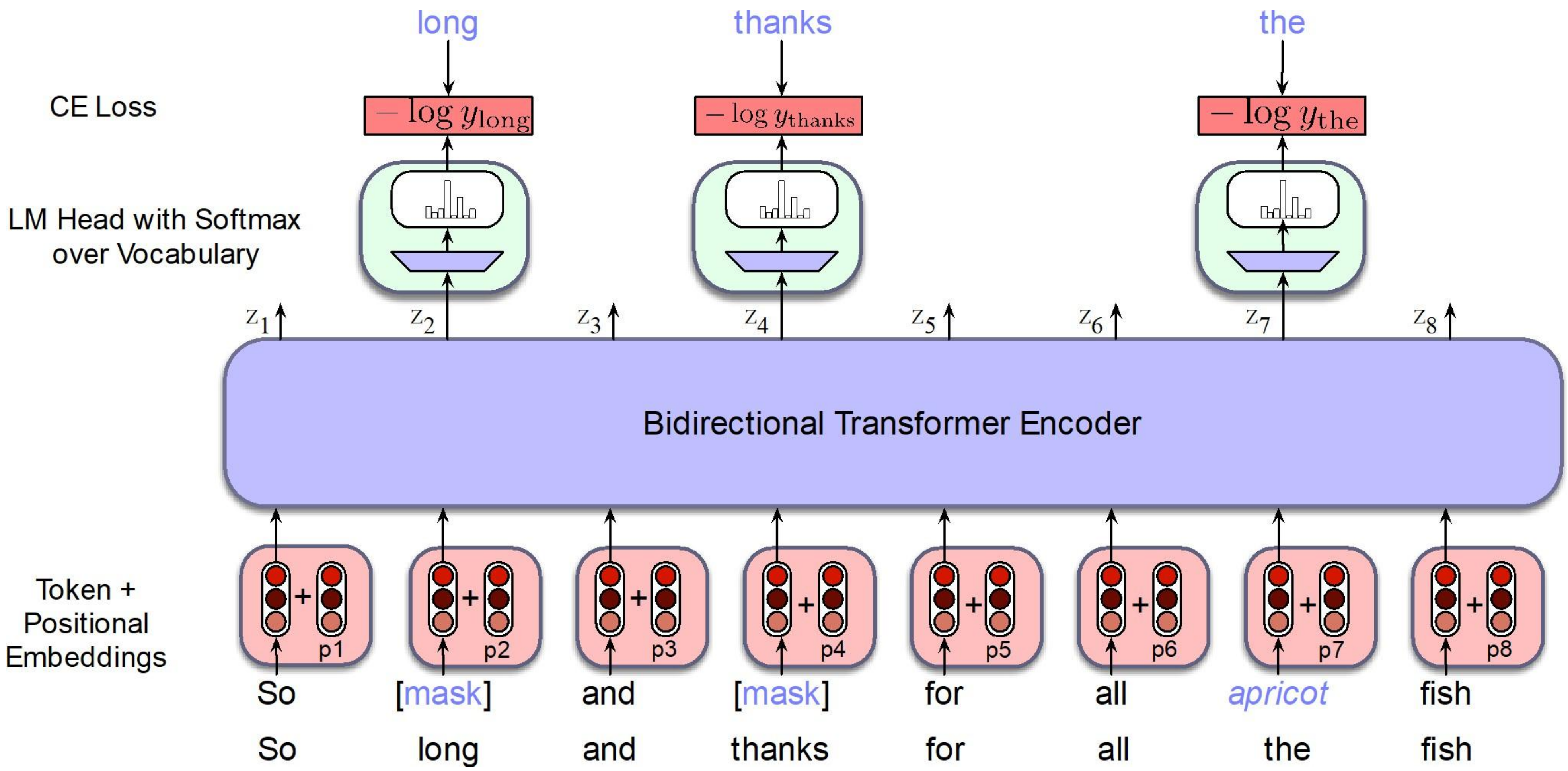
- 10% of the time, replace random word

  `went to the store → went to the running`

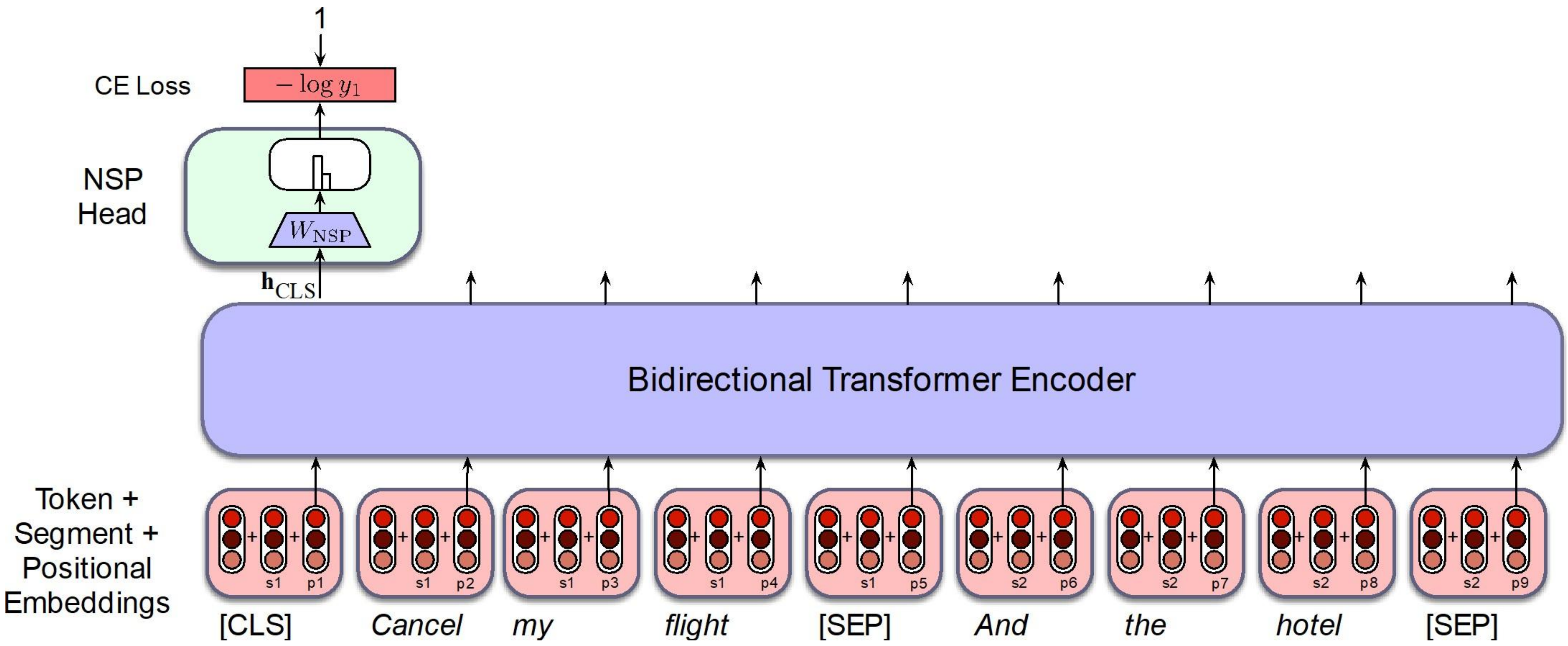
- 10% of the time, keep same

  `went to the store → went to the store`

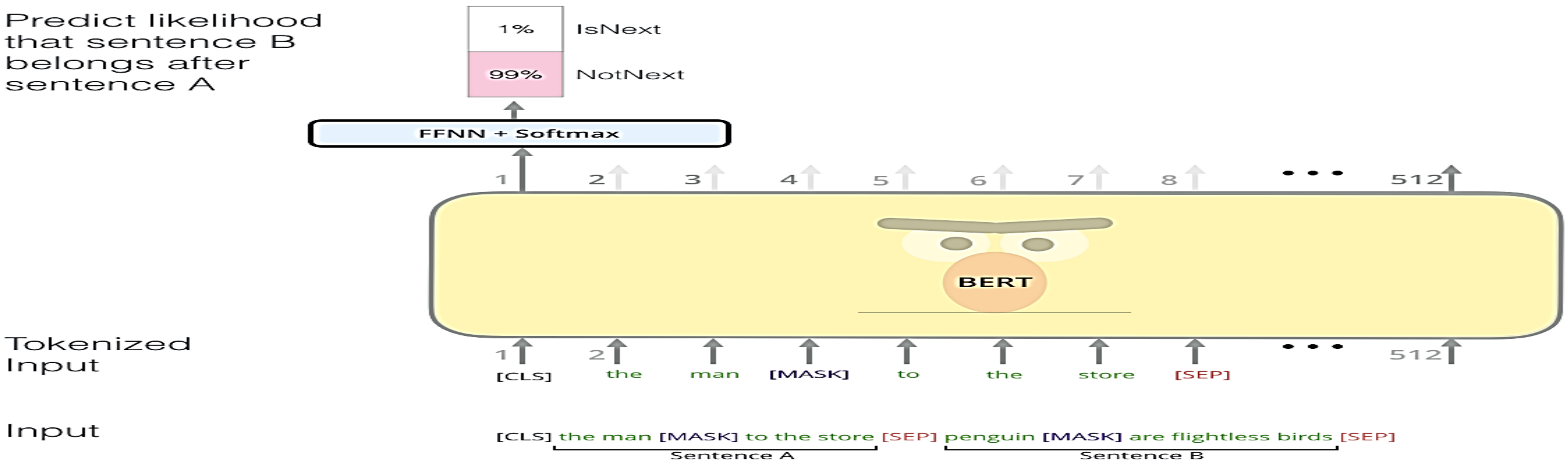
How was Bert Trained: Masked LM(MLM)



How was Bert Trained: Two-sentence Tasks



How was Bert Trained…



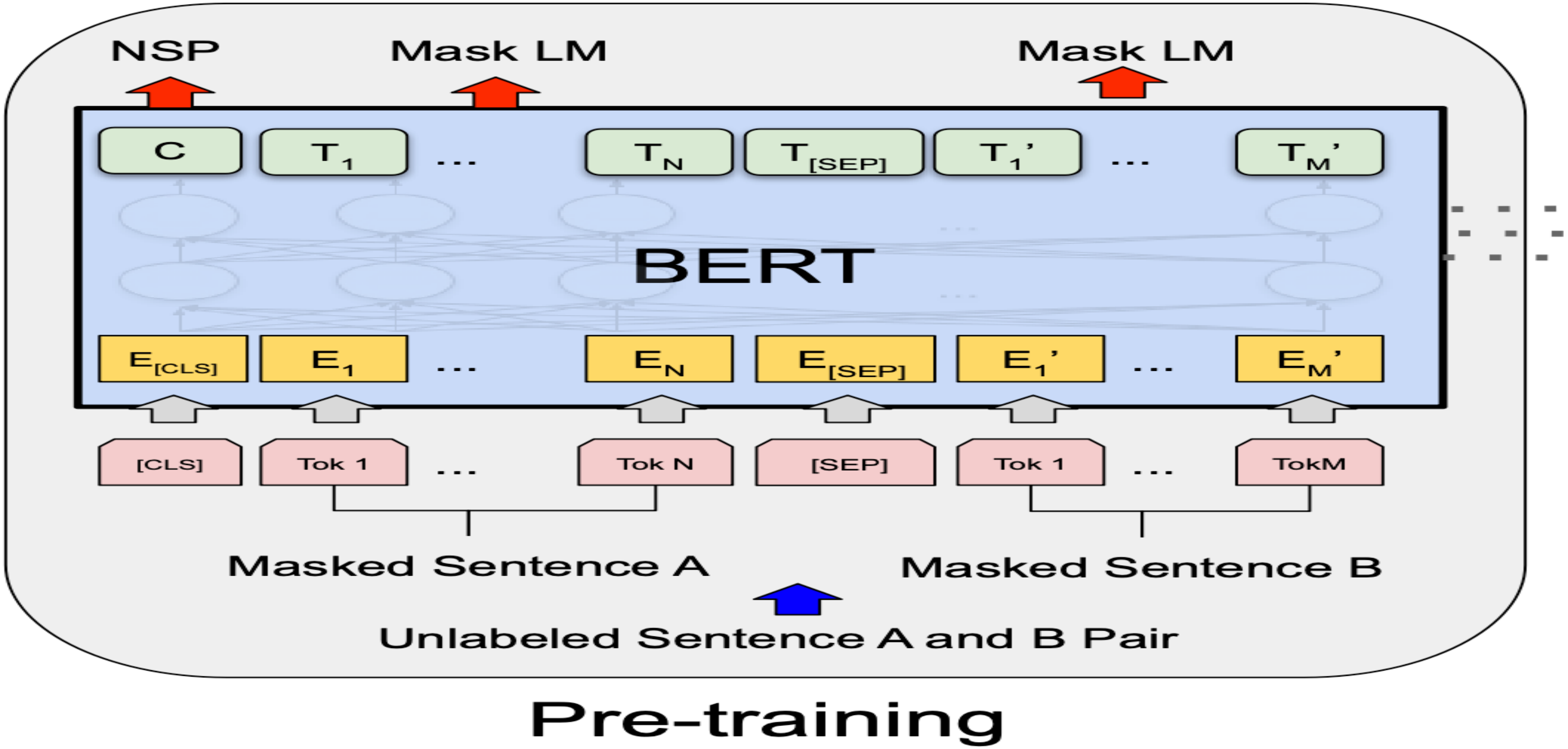
Sentence A = The man went to the store.
Sentence B = He bought a gallon of milk.
Label = IsNextSentence

Sentence A = The man went to the store.
Sentence B = Penguins are flightless.
Label = NotNextSentence

Two-sentence Tasks

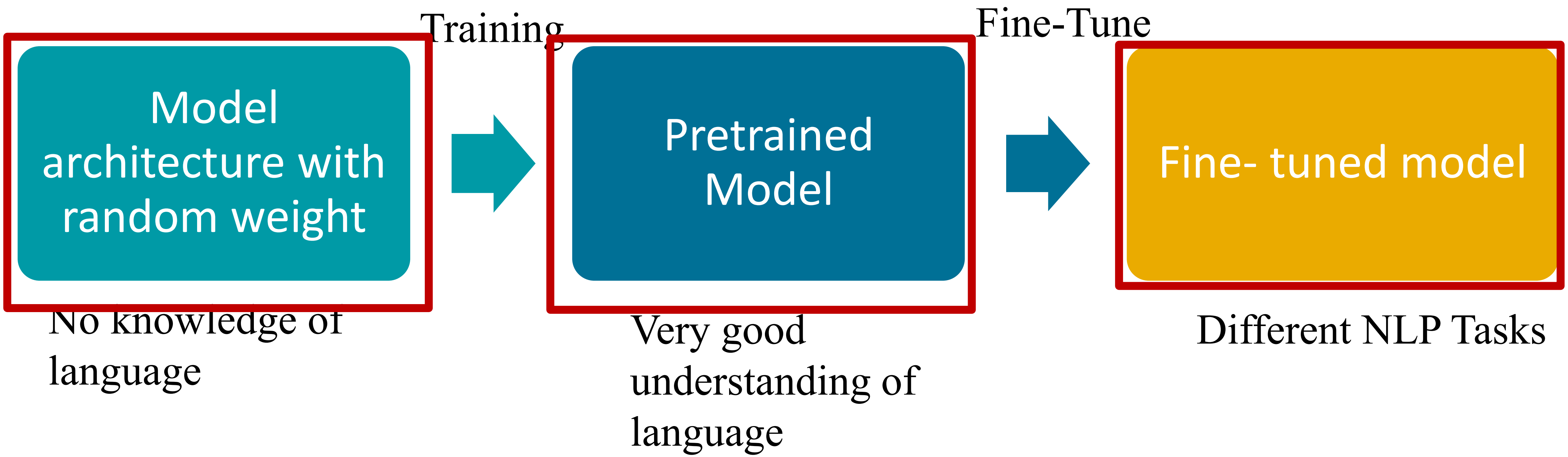
http://jalammar.github.io/a-visual-guide-to-using-bert-for-the-first-time/

BERT pre-training: Putting Together



$$\text{Total } Loss = MLM\ Loss + NSP\ Loss$$

Transfer Learning: Quick Overview



Bert: Transfer Learning

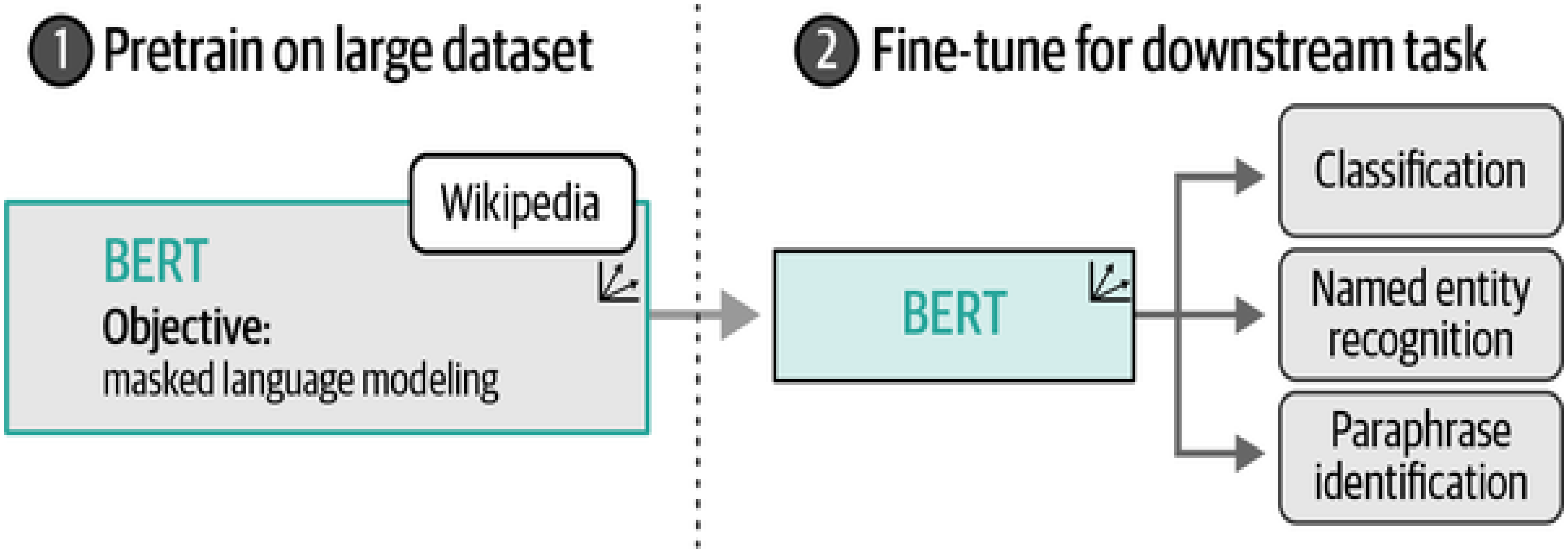
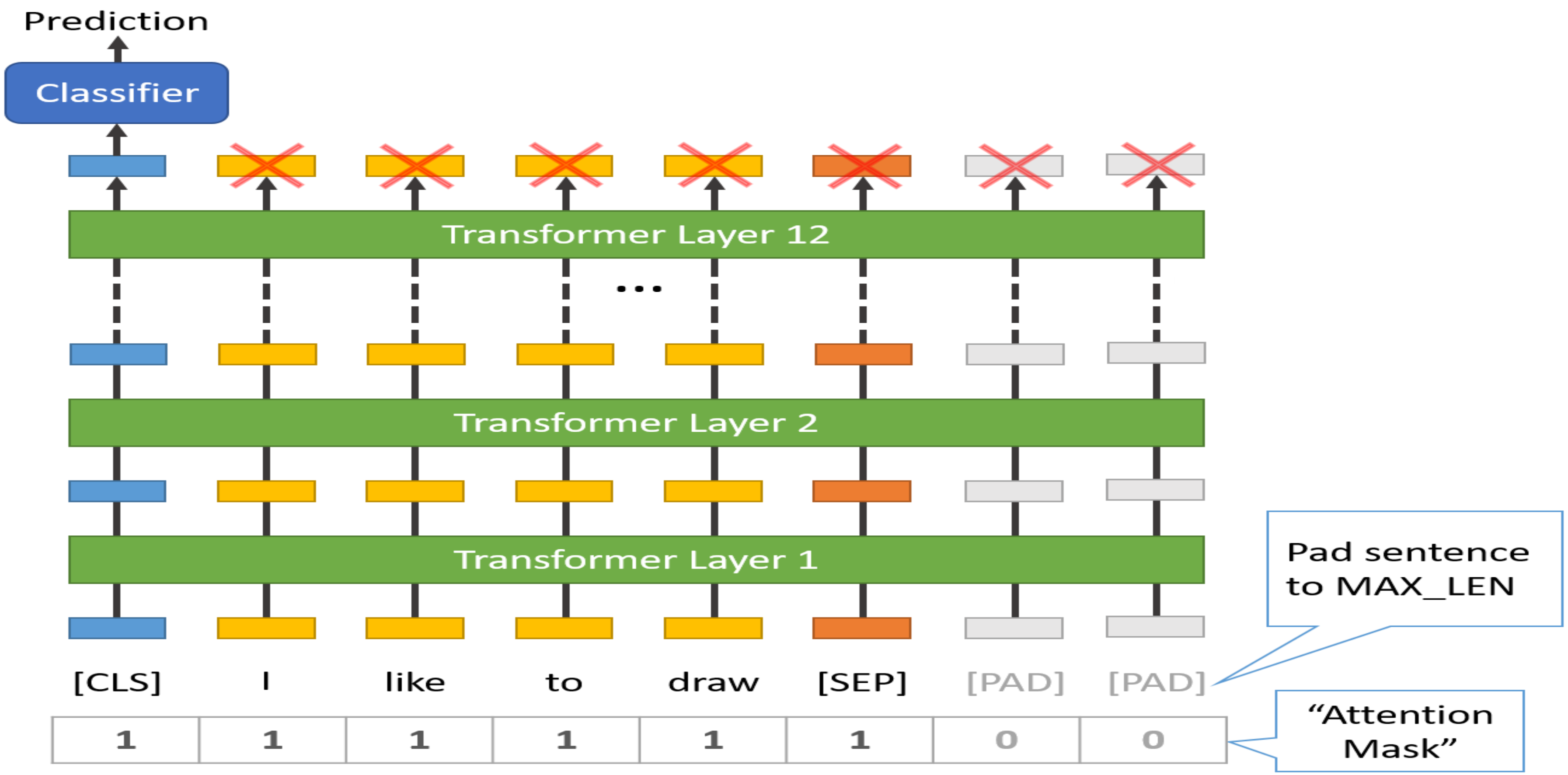


Image source: Hands-on **Large Language Models, Book**

BERT for Text Classification



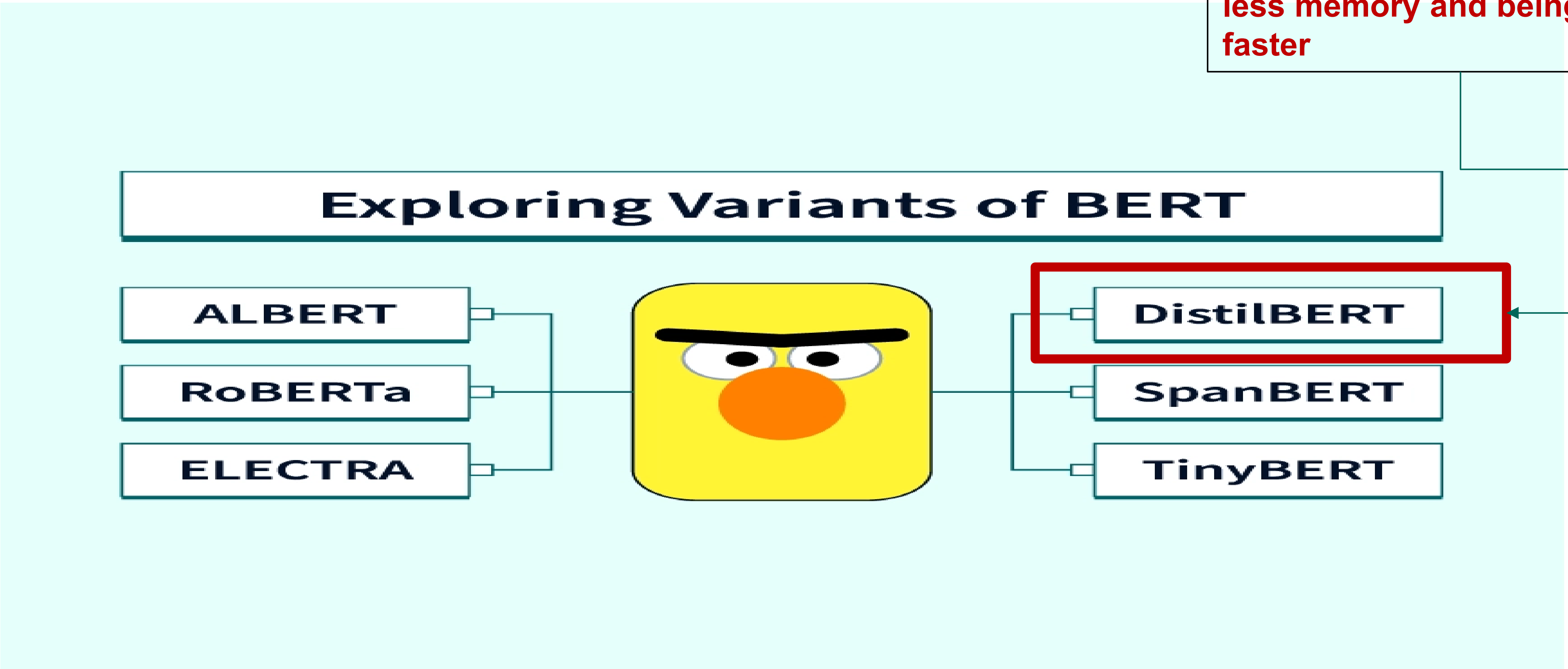
DEMO

list of the released pre-trained BERT models

BERT-Base, Uncased	12-layer, 768-hidden, 12-heads, 110M parameters
BERT-Large, Uncased	24-layer, 1024-hidden, 16-heads, 340M parameters
BERT-Base, Cased	12-layer, 768-hidden, 12-heads, 110M parameters
BERT-Large, Cased	24-layer, 1024-hidden, 16-heads, 340M parameters
BERT-Base, Multilingual Cased (New)	104 languages, 12-layer, 768-hidden, 12-heads, 110M parameters
BERT-Base, Multilingual Cased (Old)	102 languages, 12-layer, 768-hidden, 12-heads, 110M parameters

BERT VARIANTS

achieves 97% of BERT’s performance while using 40% less memory and being 60% faster



https://www.scaler.com/topics/nlp/bert-variants/

Q/A SYSTEM

Taxonomy of Q/A System

- **Information source**
- **Question types**
- **Answer type**

Information Source

- Structured Data Sources
- Databases (SQL, NoSQL)
- Unstructured Text Sources
  - Web Documents & Articles (e.g., Wikipedia, news websites), Research Papers & Scientific Literature (e.g., arXiv, PubMed): Critical for academic and medical Q/A systems.
  - Product Manuals & Documentation.
  - Books & Digital Libraries (e.g., Project Gutenberg, Google Books)
- Conversational Data Sources
  - Customer Support Logs & Chat Transcripts
  - Community Forums & Q/A Sites (e.g., Stack Overflow, Quora
  - Social Media Feeds (e.g., X): For real-time Q/A on trends, events, and opinions.

Types of Questions in Modern Systems

- Factoid Questions
- Open domain Questions
- Closed domain Questions
- Complex (narrative) Questions

Answer type

- **Extractive Answers** (Span-Based Answers)
- **Abstractive (Generative) Answers**.
- **Factoid Answers(** Knowledge-based)

  The system provides short factual answers, such as names, dates, numbers, or locations.

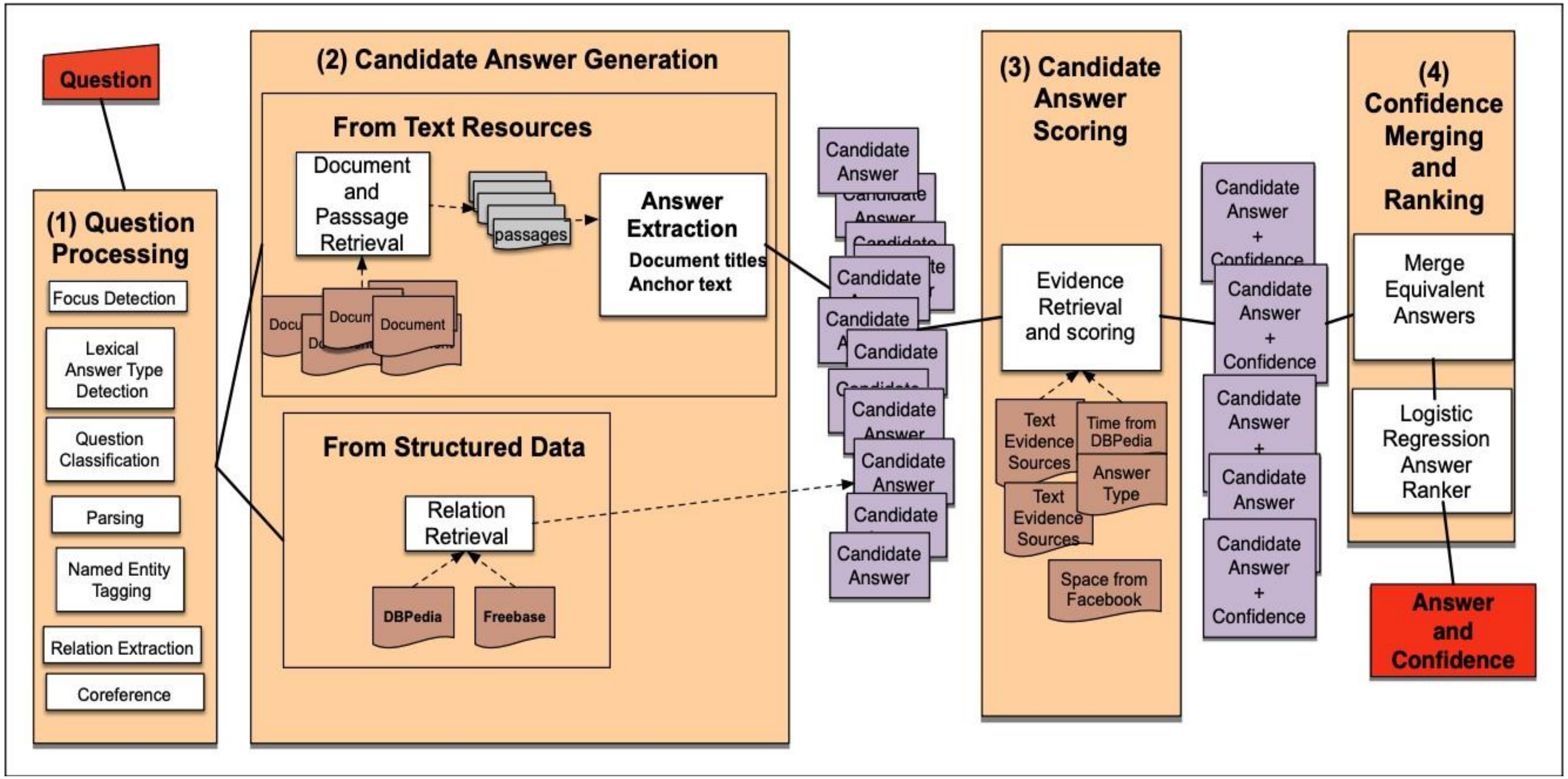
Question Answering Paradigms

- Extractive QA (SQuAD, BERT-based models)
- Knowledge-based QA
- Hybrid approaches QA
- Generative QA
- Retrieval-Augmented QA (RAG)

2011: IBM Watson beat Jeopardy champions

IBM Watson architecture

won Jeopardy on February 16, 2011!



Reading Ccomprehension (extractive QA)

Reading comprehension = comprehend a passage of text and answer questions about its content (P,Q) → A

Tesla was the fourth of five children. He had an older brother named Dane and three sisters, Milka, Angelina and Marica. Dane was killed in a horse-riding accident when Nikola was five. In 1861, Tesla attended the "Lower" or "Primary" School in Smiljan where he studied German, arithmetic, and religion. In 1862, the Tesla family moved to Gospić, Austrian Empire, where Tesla's father worked as a pastor. Nikola completed "Lower" or "Primary" School, followed by the "Lower Real Gymnasium" or "Normal School."

Q: What language did Tesla study while in school?

A: German

Slide credit: Dan Jurafsky

Stanford question answering dataset (SQuAD)

- 100k annotated (passage, question, answer) triples
- Passages are selected from English Wikipedia, usually 100~150 words.
- Questions are crowd-sourced.
- Each answer is a short segment of text (or span) in the passage.
- SQuAD remains the most popular reading comprehension dataset.

In meteorology, precipitation is any product of the condensation of atmospheric water vapor that falls under **gravity**. The main forms of precipitation include drizzle, rain, sleet, snow, **graupel** and hail... Precipitation forms as smaller droplets coalesce via collision with other rain drops or ice crystals **within a cloud**. Short, intense periods of rain in scattered locations are called "showers".

What causes precipitation to fall?
gravity

What is another main form of precipitation besides drizzle, rain, snow, sleet and hail?
graupel

Where do water droplets collide with ice crystals to form precipitation?
within a cloud

Stanford Question Answering dataset (SQuAD)

Evaluation: **E**xact **M**atch EM (0 or 1) and **F1** (partial credit).

- For development and testing sets, **3 gold answers are collected.**
- compare the predicted answer to each gold answer and take max scores.
- Take the average of all the examples for both exact match and F1.

Q: What did Tesla do in December 1878?

Answer: {left Graz, **left Graz ans**, **left Graz and severed all relations with his family**}

Prediction: {left Graz and severed}

Exact Match: max{0, 0, 0} = 0

F1: max{0.67, 0.67, 0.61} = 0.67

Neural Models For Reading Comprehension

Problem formulation

- Input: $C = (c_1, c_2, \ldots, c_N), Q = (q_1, q_2, \ldots, q_M), c_i, q_i \in V$
- Output: $1 \leq \text{start} \leq \text{end} \leq N$

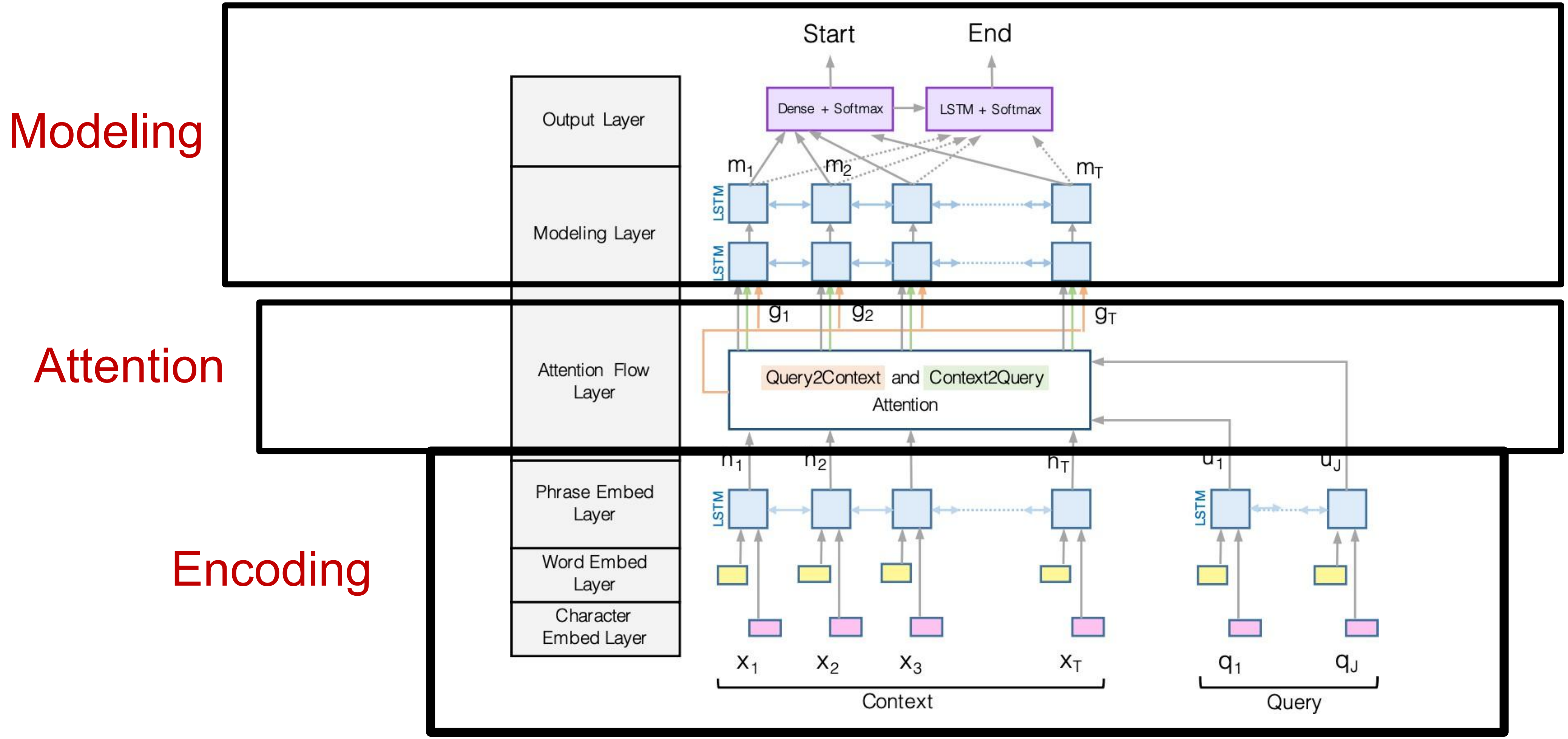
M<N

answer is a span in the passage

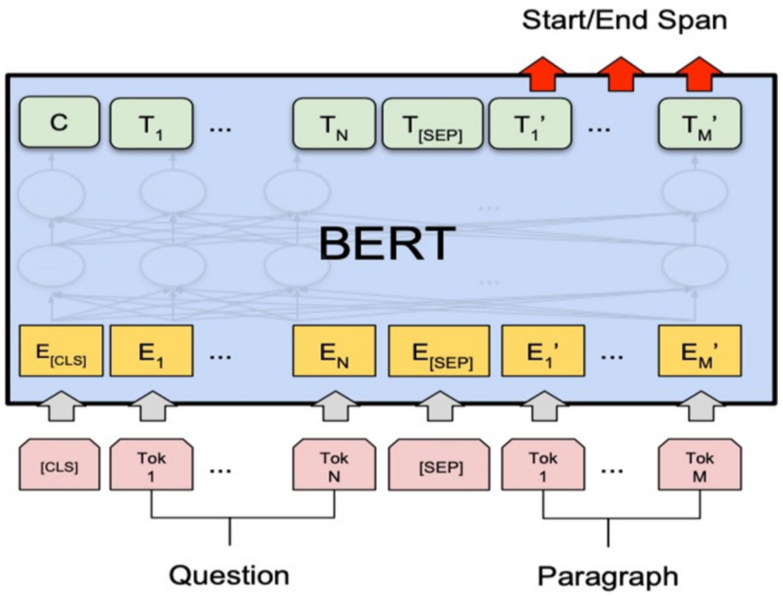
- A family of **LSTM-based models** with attention (2016–2018)
- Fine-tuning **BERT-like models** for reading comprehension (2019+)

BiDAF: the Bidirectional Attention Flow model

Exact Match (EM): 71.3%F1 score: 81.2%



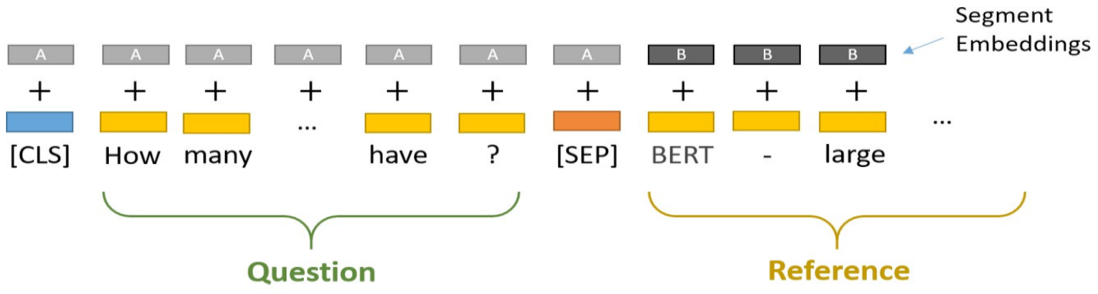
BERT for Reading Comprehension…



Question = Segment A

Passage = Segment B

Answer = predicting two endpoints in segment B



Question: How many parameters does BERT-large have?

Reference Text: BERT-large is really big... it has 24 layers and an embedding size of 1,024, for a total of 340M parameters! Altogether it is 1.34GB, so expect it to take a couple minutes to download to your Colab instance.

Image credit: https://mccormickml.com/

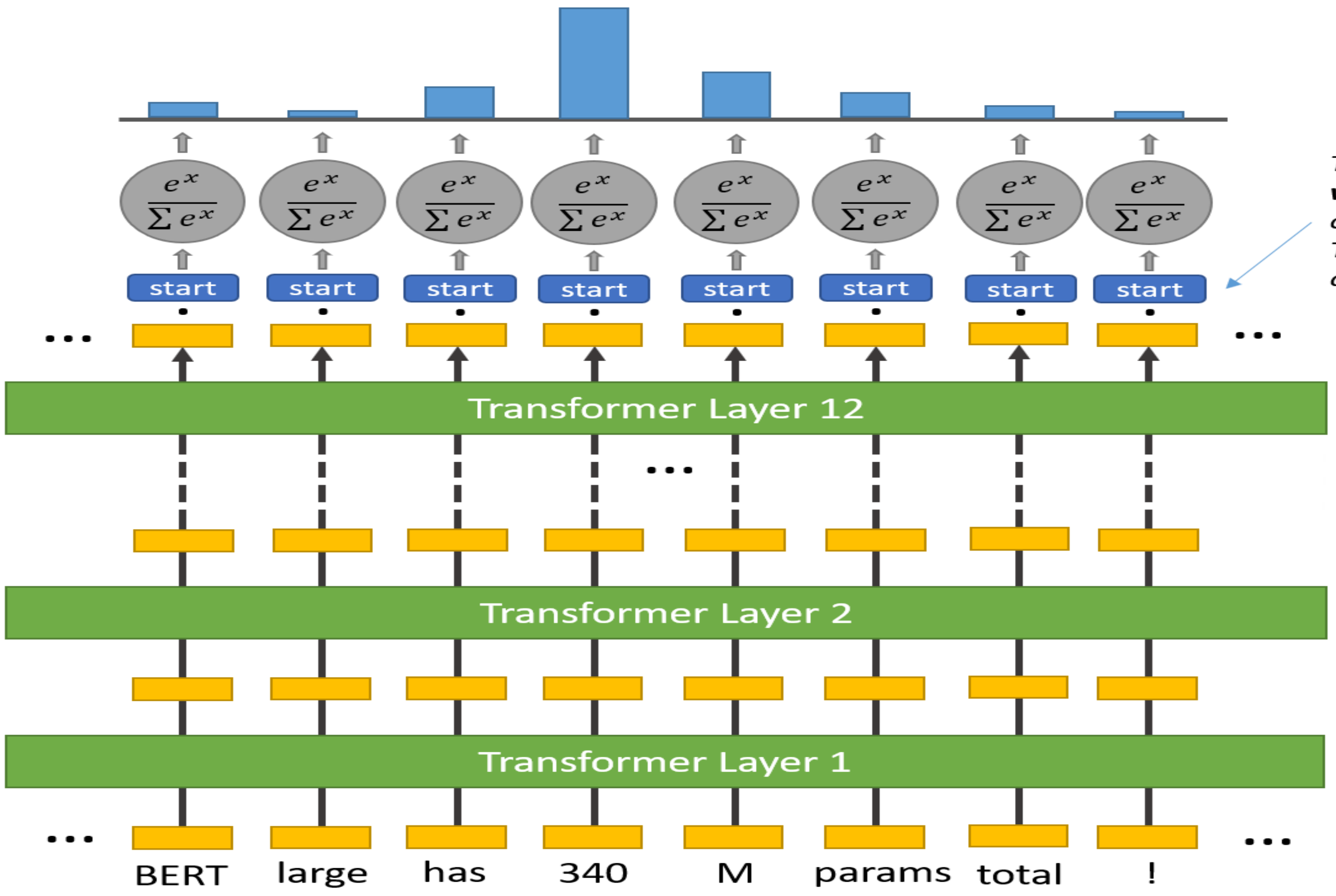
$$L = -\log p_{start}(s^{*}) - \log p_{end}(e^{*})$$

$$p_{start}(i) = \text{softmax}_i(\mathbf{w}_{start}^{\top}\mathbf{H})$$

$$p_{end}(i) = \text{softmax}_i(\mathbf{w}_{end}^{\top}\mathbf{H})$$

where $\mathbf{H} = [\mathbf{h}_1, \mathbf{h}_2, \ldots, \mathbf{h}_N]$ are the hidden vectors of the paragraph, returned by BERT

BERT for Reading Comprehension: Predict start



*This length 768 vector is the **weights** for the start token classifier.*
*The **same weights** are applied to **every position**.*

Image credit, Chris McCormick

BERT for Reading Comprehension: Predict end

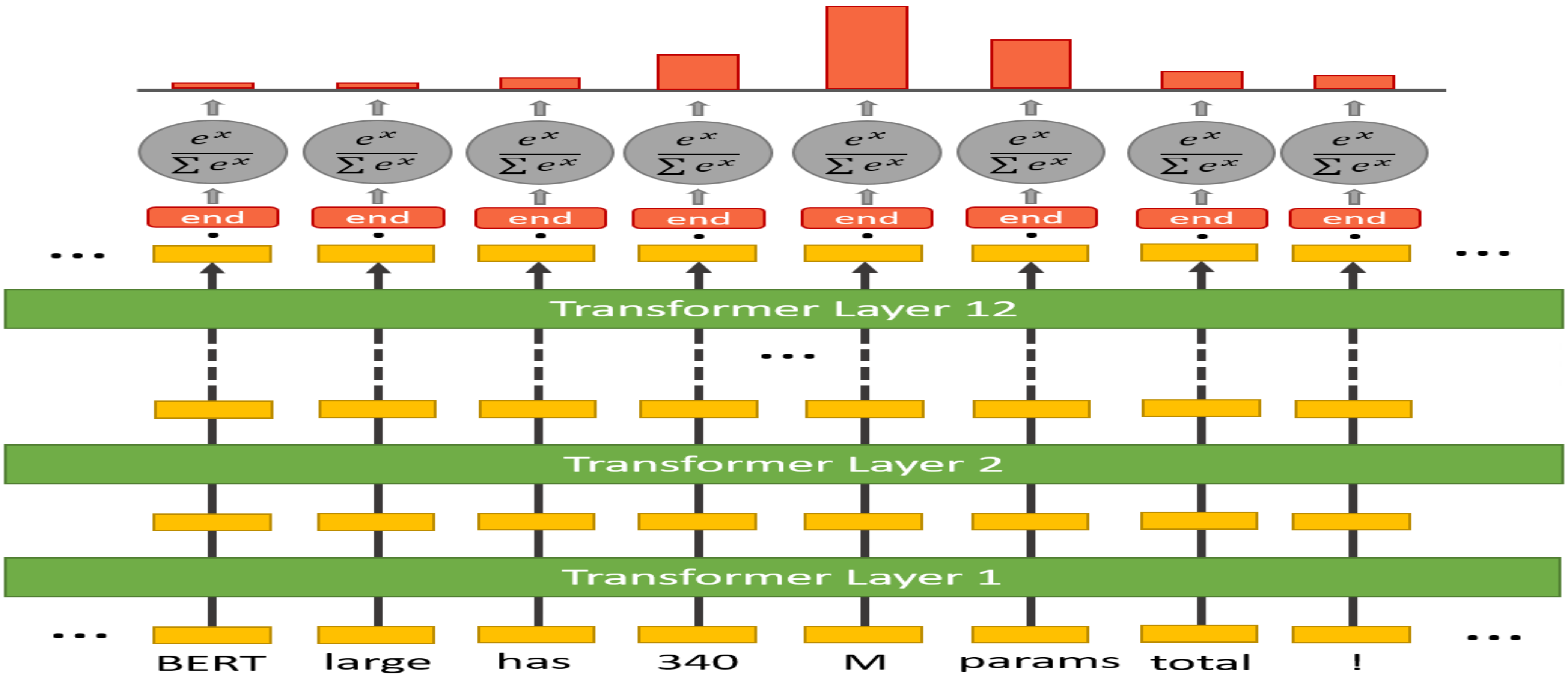


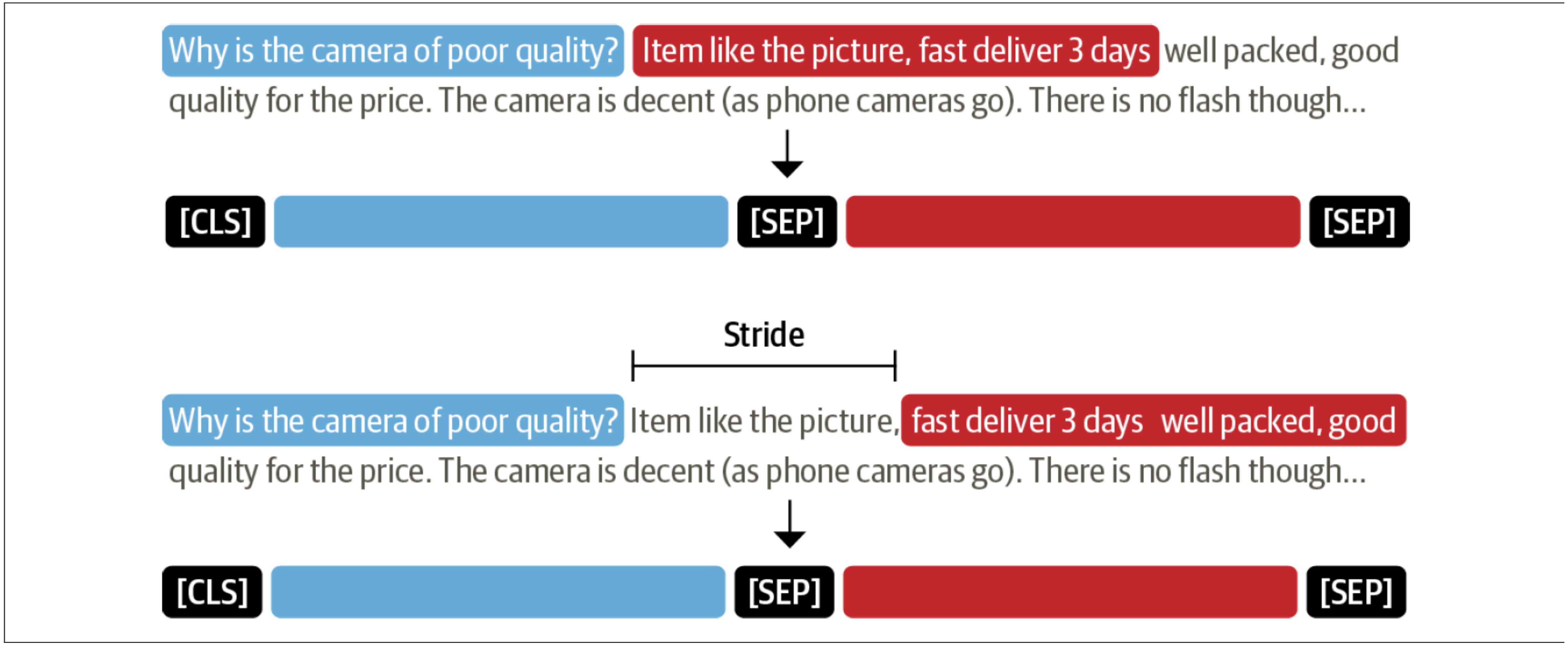
Image credit, Chris McCormick

Comparisons between BiDAF and BERT models on SQuAD 2.0

	F1	EM
BiDAF	77.3	67.7
BERT-base	88.5	80.8
BERT-large	90.9	84.1
XLNet	94.5	89.0
RoBERTa	94.6	88.9
ALBERT	94.8	89.3

Source: https://rajpurkar.github.io/SQuAD-explorer/

Dealing With Long Passages



Tokenizing Questions and Contexts for BERT-based Question Answering

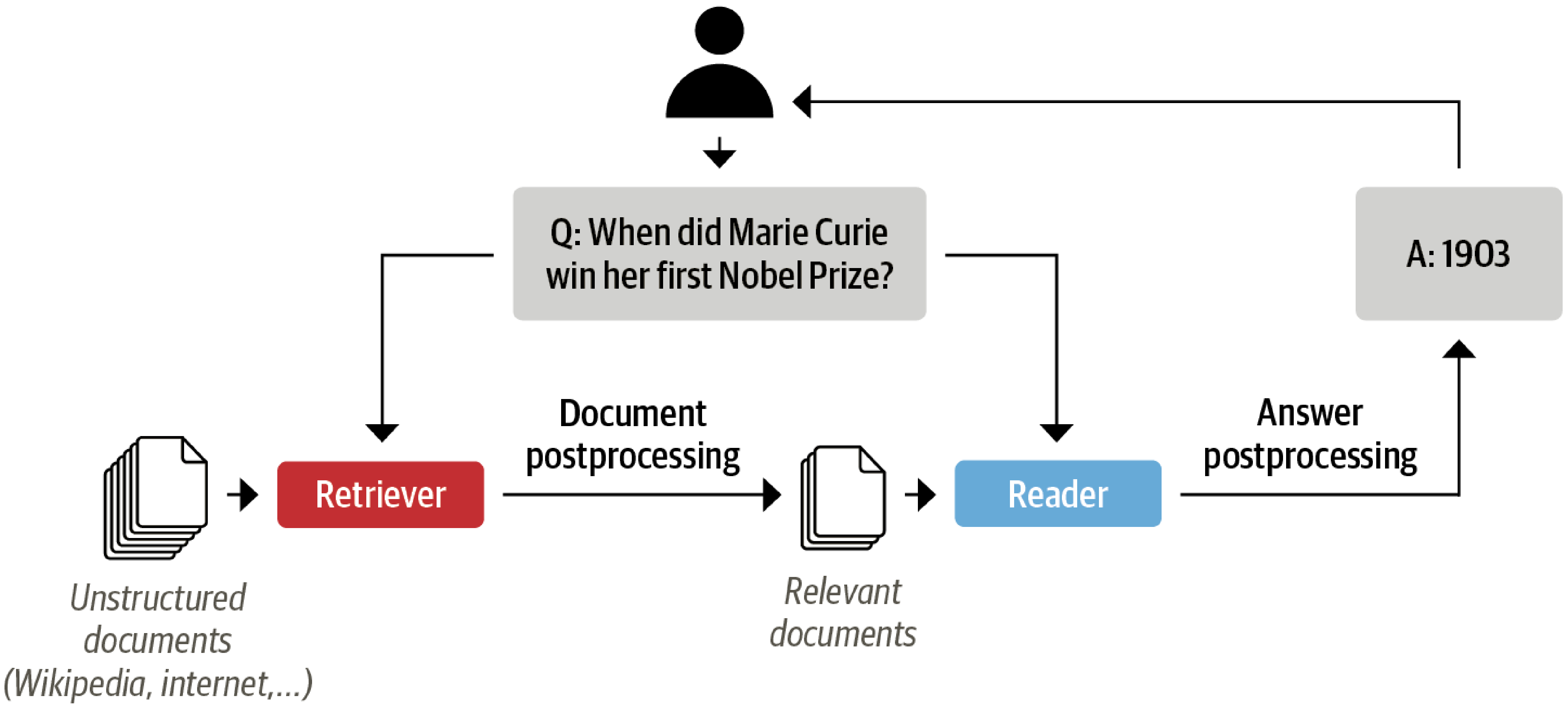
```
inputs = tokenizer(
    examples["question"],
    examples["context"],
    max_length=500,
    truncation="only_second",
    stride=25,
    return_overflowing_tokens=True,
    return_offsets_mapping=True,
    padding="max_length",
    return_tensors="pt",
    return_attention_mask=True,
    add_special_tokens=True
)
```

DEMO

Open Datasets available for Question Answering

- ***S**tanford **Qu**estion **A**nswering **D**ataset (SQuAD)*
- **WikiQA** dataset
- The **TREC-QA** dataset
- **NewsQA** dataset
- Google (NQ) dataset,7

Retriever-Reader Architecture



Retrieval document stores

	In memory	Elasticsearch	FAISS	Milvus
TF-IDF	Yes	Yes	No	No
BM25	No	Yes	No	No
Embedding	Yes	Yes	Yes	Yes
DPR	Yes	Yes	Yes	Yes

Embeddings in Information Retrieval

- Word2Vec
- GloVe
- BERT

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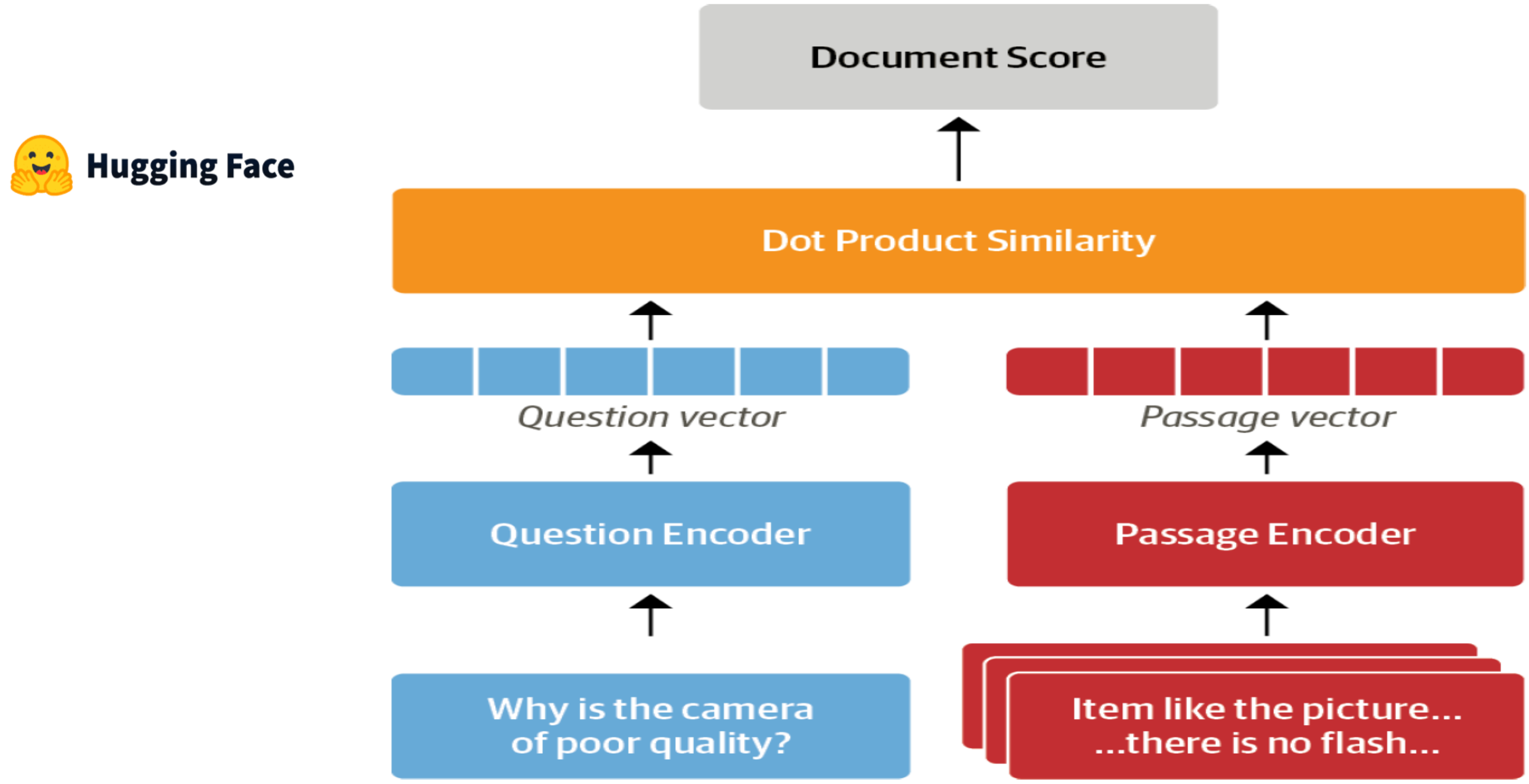
Dense Passage Retrieval (DPR)

- **Train a separate encoder** for both queries (questions) and passages (documents) to optimize their embeddings for retrieval tasks.
  - **Dual Encoder Architecture**
  - **End-to-End Training**

Dense Passage Retrieval(DPR)



Haystack library

- developed by deepset
- based on the retriever-reader architecture
- abstracts much of the complexity
- integrates tightly with Transformers.
  - Document store
  - Pipeline

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DEMO

other frameworks similar to Haystack

- DeepPavlov
- DrQA

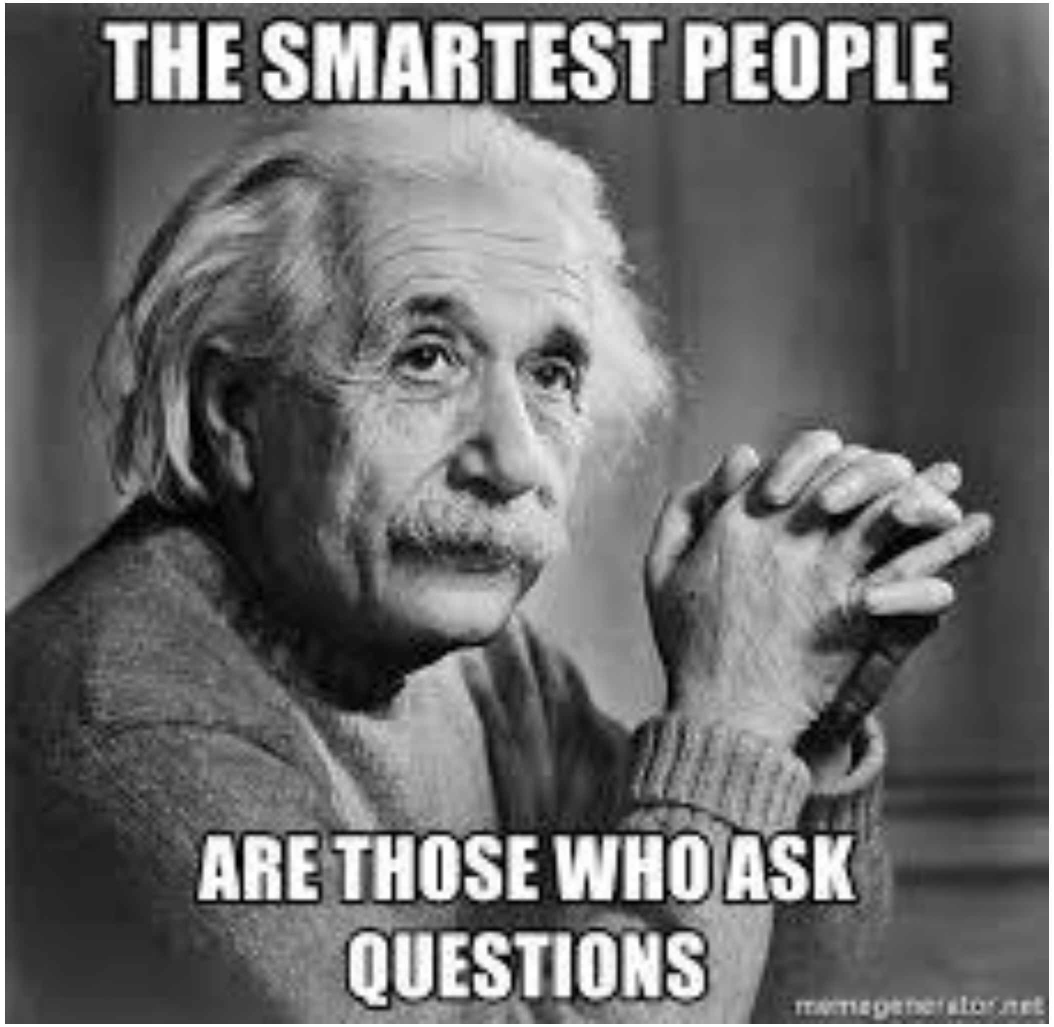
Evaluating the Reader

- *Exact Match (EM)*
- *F1-score*

Going Beyond Extractive QA

- Retrieval-augmented Generation(RAG)
  - Based on LLM

Q&A


THE SMARTEST PEOPLE
ARE THOSE WHO ASK QUESTIONS
memegenerator.net